Renegade Stick Assembly Guide

Step-by-Step Assembly Instructions with Images

Picture 1 - the Renegade Stick

License

Please be careful, especially with Li-Po batteries—they can be dangerous if mishandled. Take your time and stay safe!

Designed by Petros Matsakos pmatsakos@gmail.com

Dedicated to

my best friend Nikos for the inspiration

Prusa Research for the open-source open-hardware spirit

Renegade Stick Assembly Guide v1.0 - December 2025

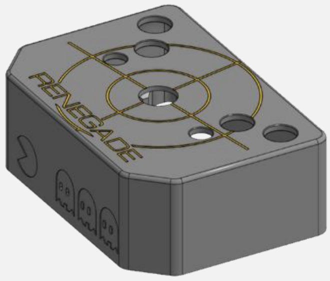
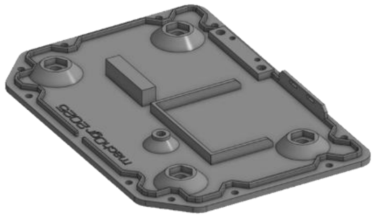
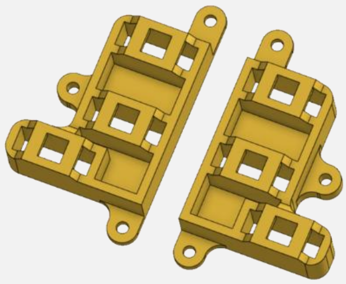
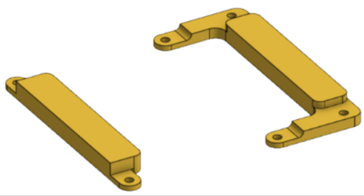
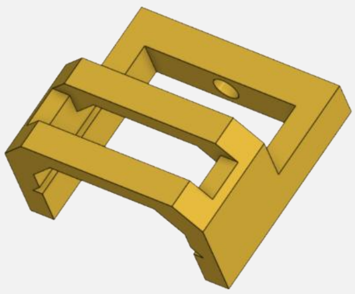
Introduction

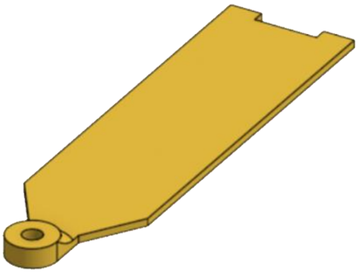
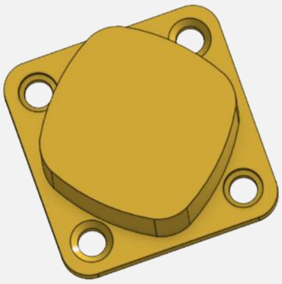
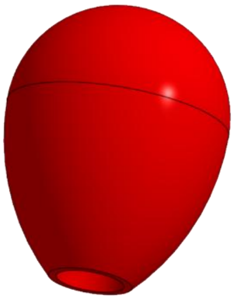
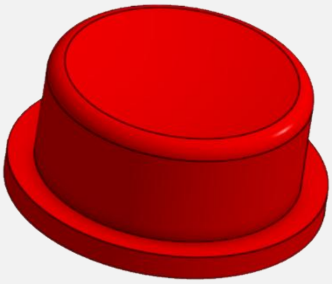
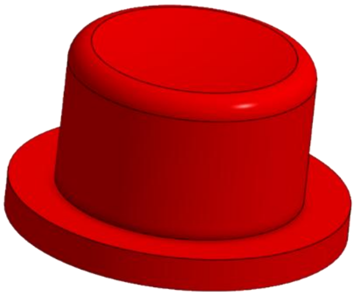
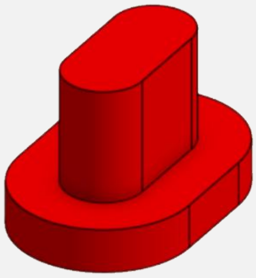
This guide will walk you through building your own Renegade Stick, step by step. We'll cover everything from 3D printed parts to electronics and screws—no experience needed!

Table of Contents

1. 3D printed Parts List

Hereby you can find the 3D printed parts list required.
All the 3D files are ready for you on printables.com—just pick your favorite color and start printing!

Part Name	Quantity	Description	Image
Base Box Top	1	Main case. Option available for multicolor print	
Base Box Bottom	1	Bottom case	
microswitch bracket Left & microswitch bracket Right	1 each	Bracket holding button microswitches	
Stick bracket bottom & Stick bracket top	1 each	Mounting brackets for the main stick	
microswitch side clip	1	Mounting bracket for side microswitches	

Battery holder	1	Holding clip for the Li-Po battery	
Stick restrictor	1	Optional item to restrict the stick movement to and to protect any wires from getting pinched in the way	
Joystick handle	1	To substitute the stick assembly “ball” type handle with something more “classic “	
Button Big	4	You will need 4 of these	
Button Small	2	You will need 2 of these	
Button Side	2	You will need 2 of these	

2. Non 3D printed parts

Hereby you can find the list of non-3D printed required for the build.
You can source these parts yourself or have a look on the [renegade-stick on github] page for some suggested online resources. Make sure you double check the quantities required !

Part Name	Quantity	Description	Image
M3x3.0 or M3x4.0 Heat set inserts Short Version	21	You need the short version either with 3.0 or 4.0 depth	
Screw M3x8mm DIN965 (Countersunk)	10	Prefer Black ones if you print a black colored bottom. Alternatively you could use equivalent countersunk screws with a torx or allen head	
Screw M3x8mm DIN912 (Socket Head)	11	You can use any M3 non countersunk as long as it is 8mm screw body long	
Microswitch Mini SPDT ON-(ON) - without Lever	8	prefer the no lever version	
Suction cup with M6 screw	4	Suction cup with a M6 screw.	

Nut M6 DIN934	4	Used for securing the suction cups above	
Arcade Joystick Assembly	1	There is no specific brand for this. There are many similar assemblies on the market. Pay attention to the photo to source the exact one. Mind you the mini switches at the bottom.	
Batt 3.7V 2500mAh	1	You need one with JST PH2.0 connector. You can easily source this if you search for LiPo 555060	
Screw Terminal 6P 2.54mm	4	You can buy assortment for 2pin-3pin-4pin-5pin-6pin screw terminals but in most cases you'll be ok with just buying 6pin and laying them down as needed. Make sure they are the 2.54mm pitch pin variant	
Hook-Up Wire 22 to 24 AWG 0.32 to 0.20 mm2 Assortment colors (Stranded		Prefer different colors so to make your life easier identifying which switch to which board pin. Prefer stranded type Diameter size is a matter of preference here	
FireBeetle2 ESP32-C6 (DFR1075)	1	Select only **ONE** of following 3 board. All have been tested to work. Prefer ESP32-C6 being cheaper, more modern and feature-rich. Pay attention to the DFR code when ordering.	

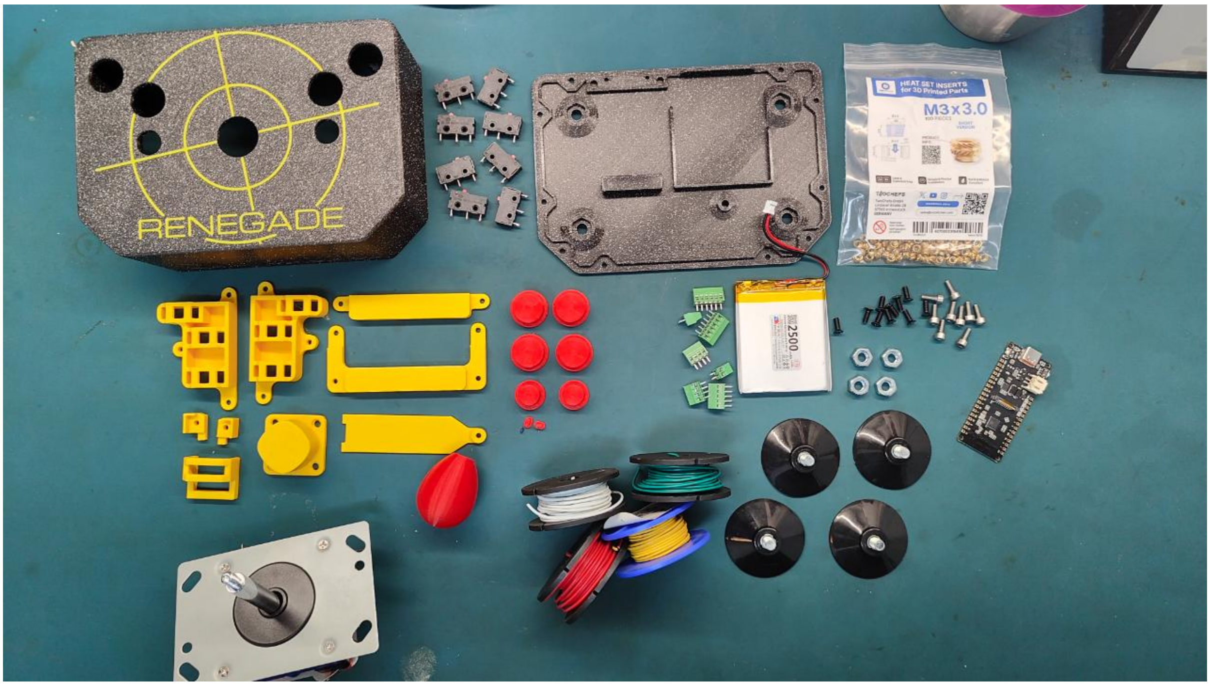
FireBeetle2 ESP32-S3(N4) (DFR1145)	1	Select only **ONE** of following 3 board. All have been tested to work. Prefer ESP32-C6 being cheaper, more modern and feature-rich. Pay attention to the DFR code when ordering.	
FireBeetle2 ESP32-E N16R2 (DFR1139)	1	Select only **ONE** of following 3 board. All have been tested to work. Prefer ESP32-C6 being cheaper, more modern and feature-rich. Pay attention to the DFR code when ordering.	
Quick Disconnect - Female Bare 2.8mm	16	**OPTIONAL item**. Only needed if you don't wish to solder the microswitch terminals. The 2.8mm version is for microswitches	
Quick Disconnect - Female Bare 4.8mm	8	**OPTIONAL item**. Only needed if you don't wish to solder the microswitch terminals The 4.8mm version is for the joystick assembly	

4. Tools Required

- Phillips Screwdriver PH0
- Flat head Screwdriver
- Allen Screwdriver Hex 2.5
- Soldering Iron (with optional Heat insert adaptor)
- Side Cutters
- Hex Key Set [Insert Image]
- Other: [Insert Image]

5. Preparation

Before starting, ensure all parts are available and 3D printed components are free from defects. Clean up any support material or rough edges on the printed parts.

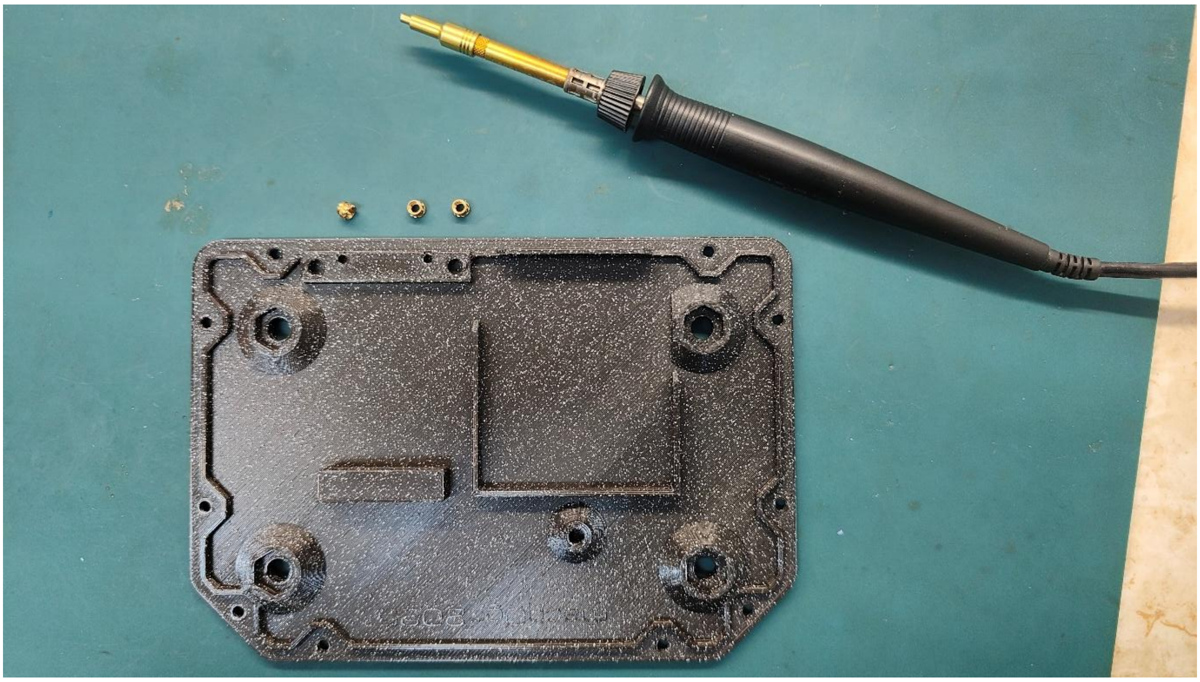


Picture 2 – parts of the Renegade Stick

6. Assembly Steps

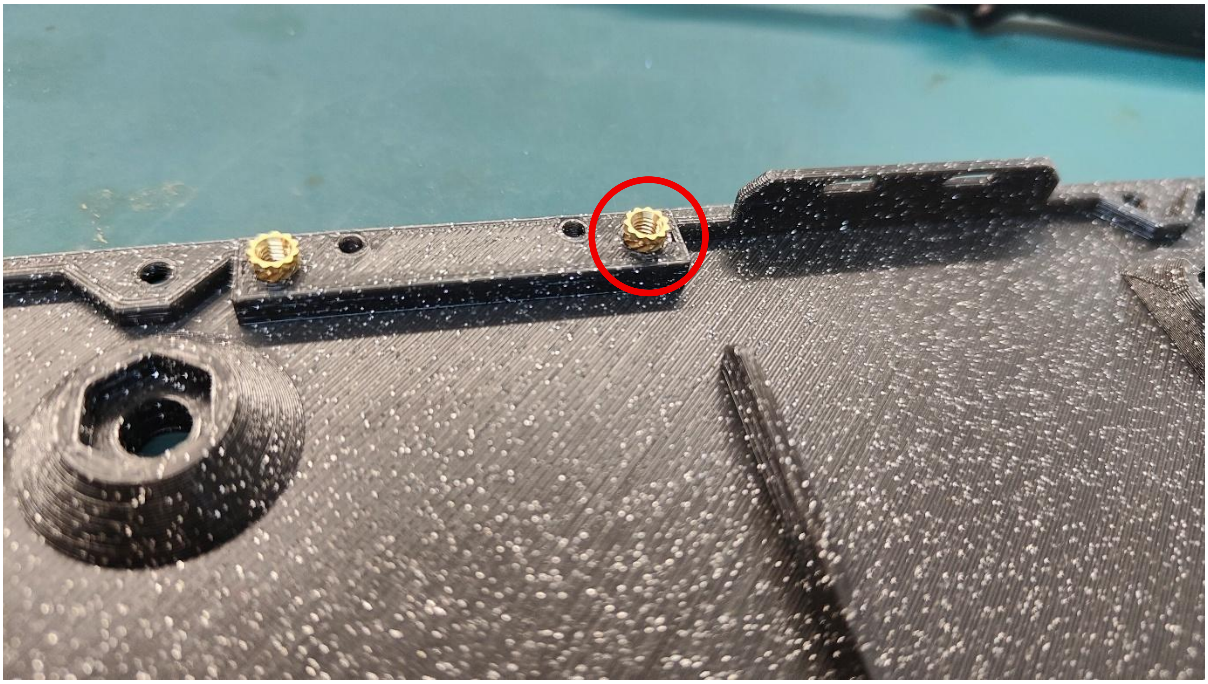
6.1 Installing your heat inserts

Start with installing the heat inserts. If you have the soldering iron heat set adaptor, you'll make your life easier.

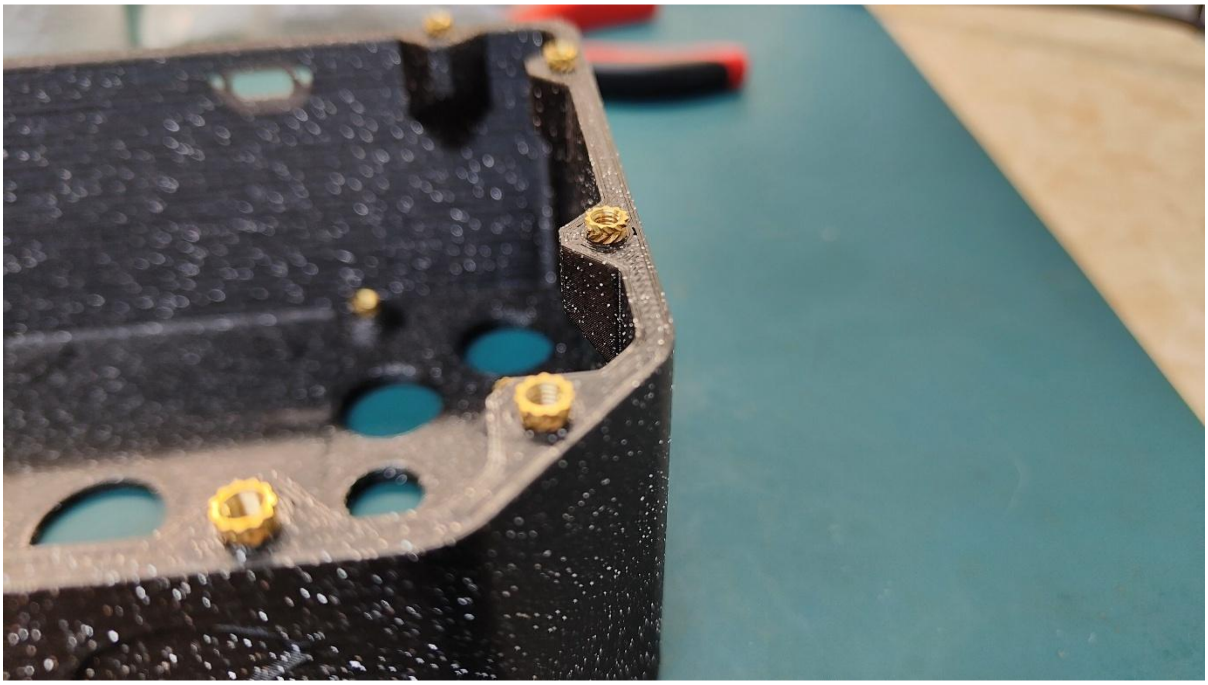


Picture 3

Start by placing all your heat inserts by hand into the relevant holes. Pushing in the bottom flanged part of the heat insert to clip into the corresponding hole.

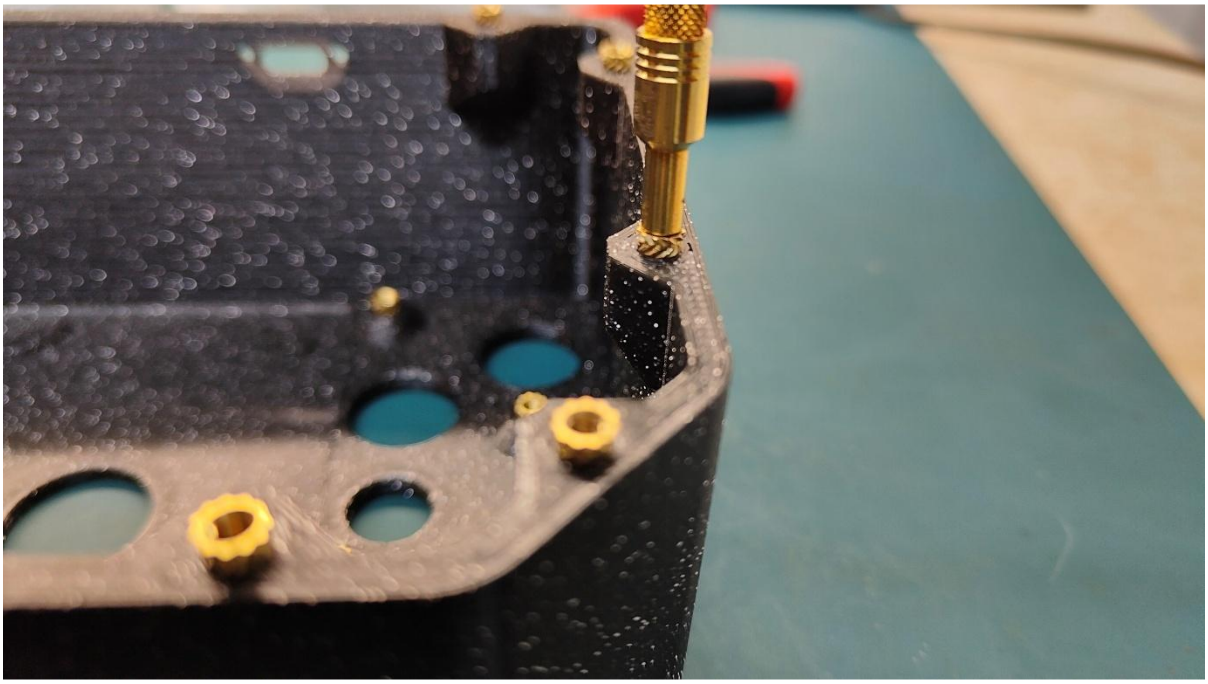


Picture 4

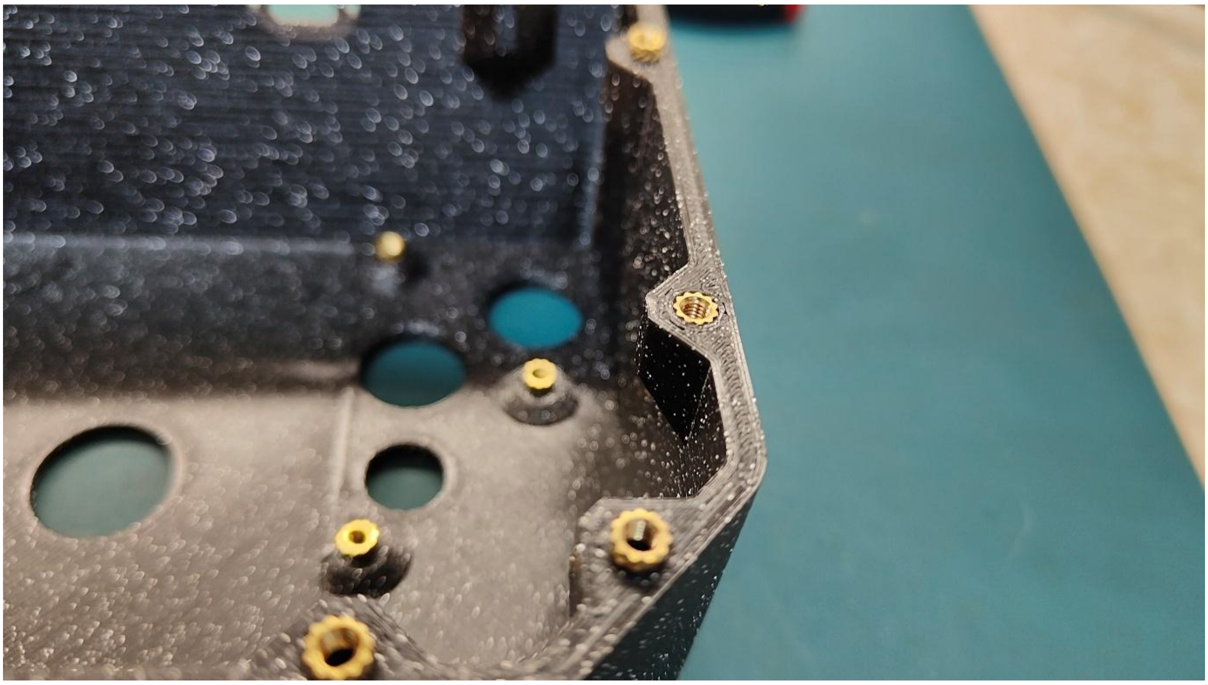


Picture 5

Heat the insert with your soldering iron or adapter. You may want to set your hot iron 10-15 degrees more than the temperature you printed the material. If you printed PLA at 200 then try setting heat iron to 215. Let the heat insert melt into the plastic on its own—no need to force it. Gravity's got your back!



Picture 6



Picture 7

6.1 Assemble the PCB board

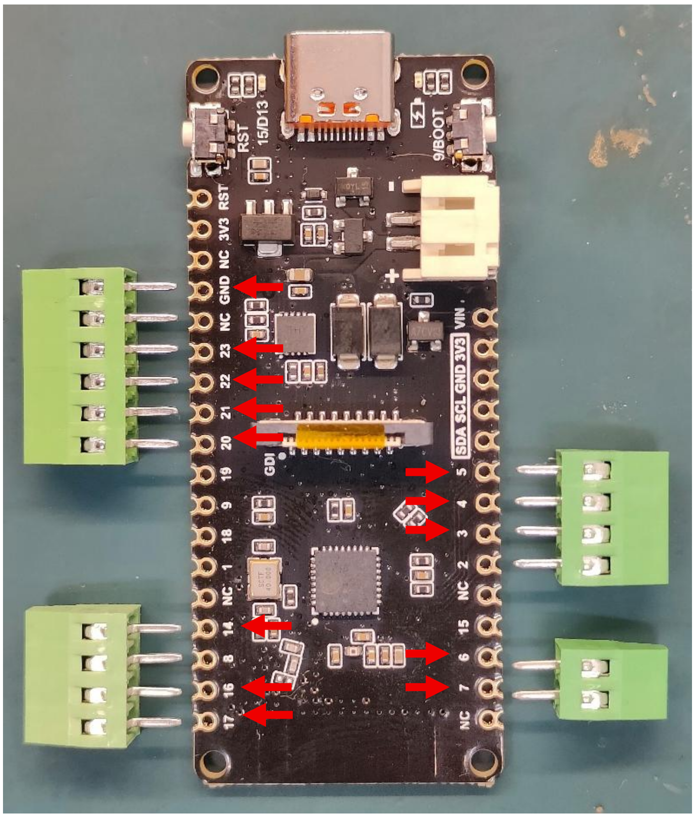
You will need to solder the PCB terminals on your PCB board. You can solder wires straight to the board, but using PCB terminals makes things much easier. Trust me, it'll save you a headache later!

Identify your board you acquired in the below matrix and have a look at the proposed connection pinouts below. You will need to install PCB terminals to cover all the proposed required pins of the board. If you want to use other pins you will have to fiddle with the code to change the mapping. If you do, mind you that not all pins can be used as inputs, I marked the ones you should not use in the comment section of the code on the board header files.

I suggest you go with the proposed pin connections below as I have tested them thoroughly in each board and I know they work well with this application.

	ESP32-C6 DFR1075	ESP32-S3 DFR1145	ESP32-E (N16R2) DFR1139
BUTTON_1	GPIO_NUM_3 (A2)	GPIO_NUM_4 (A0)	GPIO_NUM_22 (SCL)
BUTTON_2	GPIO_NUM_4 (A3)	GPIO_NUM_5 (A1)	GPIO_NUM_21 (SDA)
BUTTON_3	GPIO_NUM_5 (A4)	GPIO_NUM_6 (A2)	GPIO_NUM_15 (A4)
BUTTON_4	GPIO_NUM_14 (D3)	GPIO_NUM_8 (A3)	GPIO_NUM_12 (D13)
BUTTON_5	GPIO_NUM_16 (TX)	GPIO_NUM_10 (A4)	GPIO_NUM_4 (D12)
BUTTON_6	GPIO_NUM_17 (RX)	GPIO_NUM_11 (A5)	GPIO_NUM_17 (D10)
START_BUTTON	GPIO_NUM_6 (D12)	GPIO_NUM_9 (D7)	GPIO_NUM_13 (D7)
SELECT_BUTTON	GPIO_NUM_7 (D11)	GPIO_NUM_0 (D9)	GPIO_NUM_14 (D6)
DPAD_L	GPIO_NUM_20 (SCL)	GPIO_NUM_18 (D6)	GPIO_NUM_0 (D5)
DPAD_R	GPIO_NUM_21 (MI)	GPIO_NUM_7 (D5)	GPIO_NUM_26 (D3)
DPAD_UP	GPIO_NUM_22 (MO)	GPIO_NUM_38 (D3)	GPIO_NUM_25 (D2)
DPAD_DOWN	GPIO_NUM_23 (SCK)	GPIO_NUM_3 (D2)	GPIO_NUM_1 (TX)

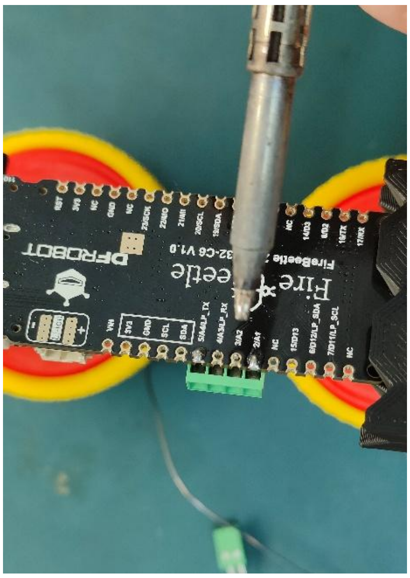
Identify your board markings from the above table and align the PCB terminals that you have so you cover(use) all the pins above. Make sure to also **include the GND pin** (on some boards, GND can be found on either side, you need to one of them, either will do)



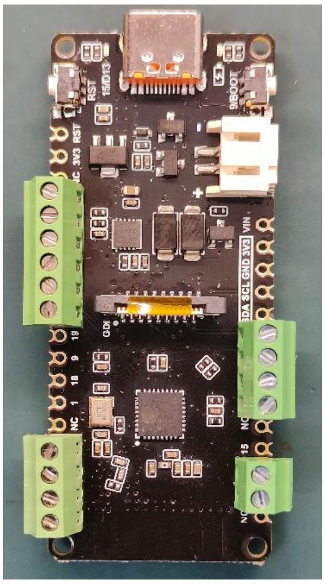
Picture 8 – marking the required pins

Take Picture 8 above as an example. This is an ESP32-C6 DFR1075 board which the needed pins are 2, 3, 4, 5, 6, 7, 16, 17, 20, 21, 22, 23, GND. I did not use the exact number of ways for the PCB terminals as you can see since these were all I had. What matters is that you cover all the pins that you'll need. In the above example I won't use pins 2, 8, NC which are not needed but my PCB terminals connect to them. When wiring you'll have to check the right ones to use.

Once you identify them solder the PCB terminals



Picture 9 – soldering the terminals

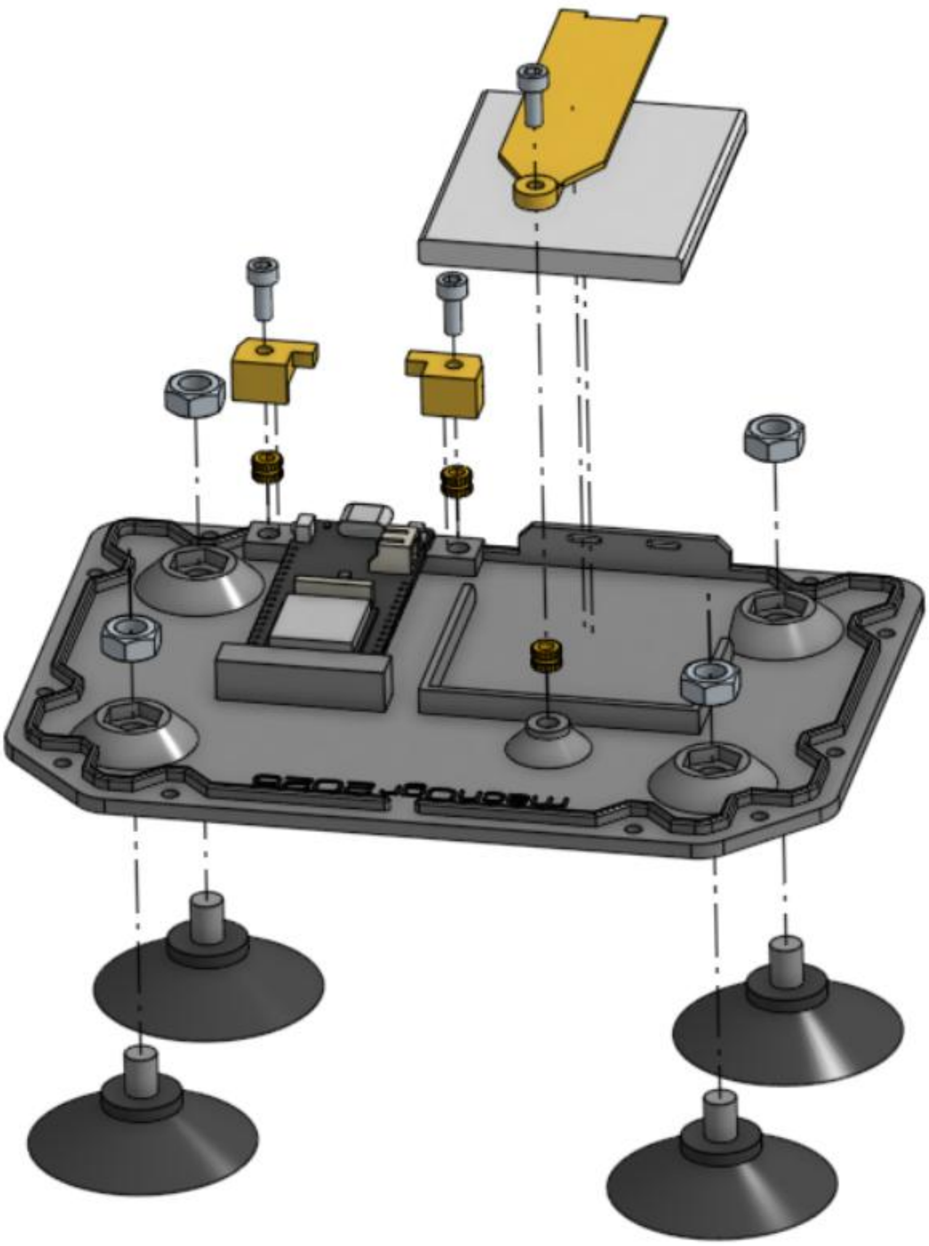


Picture 10 -board complete

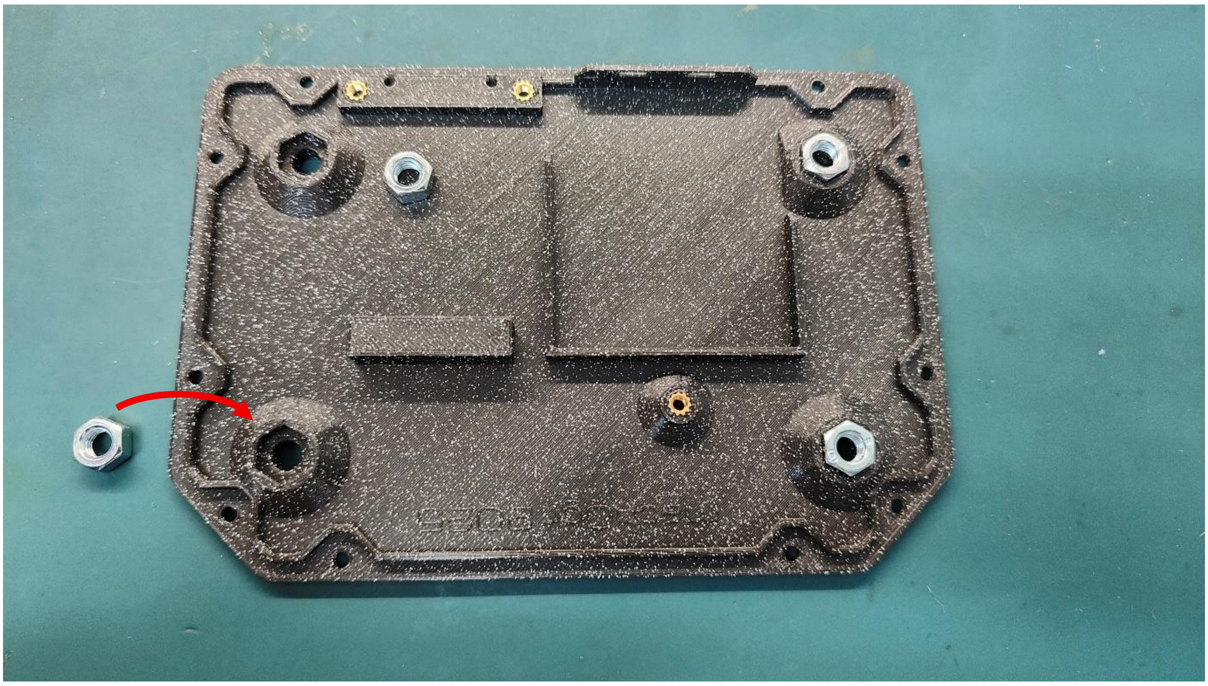
At this point, you have the option to go ahead and flash the board (visit [Renegade Stick Gitgub]() page) or flash the board after you complete the build.

6.2 Assemble bottom section

Let's assemble the bottom section. The exploded view below may help you identify the parts needed.



Start by pushing in the M6 nuts used to mount the suction cups.



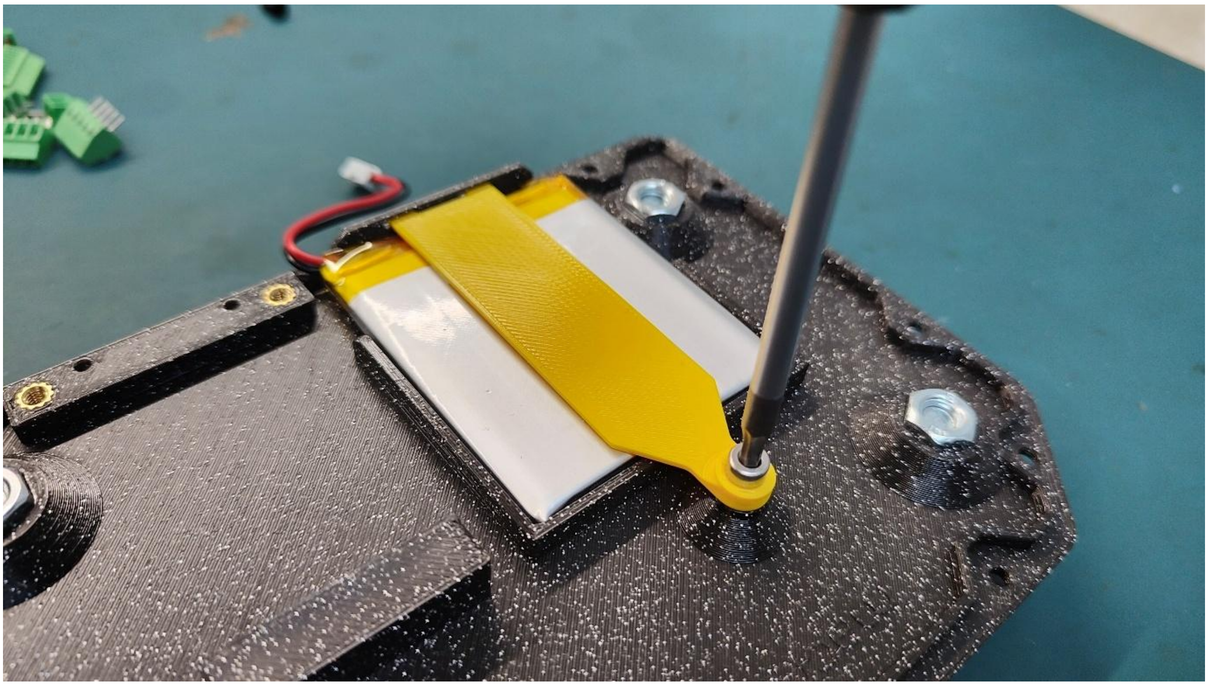
Picture 11 – insert the M6 nuts that secure the suction cups

Continue by screwing in the suction cups. No tool needed, just tighten them by hand.

Picture 12

Place the battery into its position as shown below, mind the cable side.

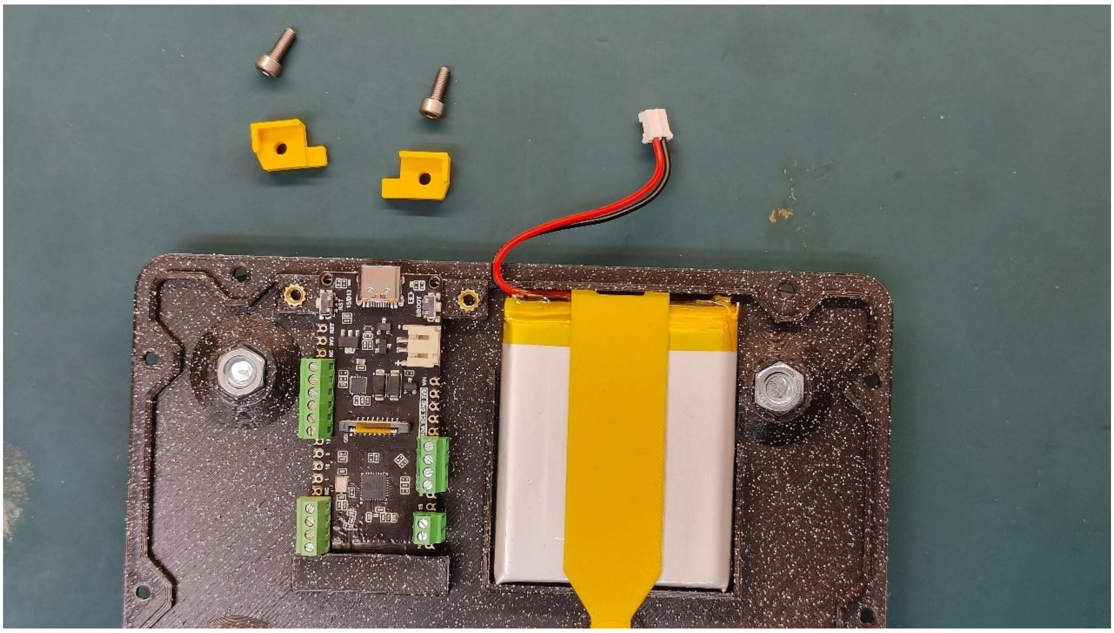
Next insert the battery clip and tighten it with an M3x8mm screw



Picture 13

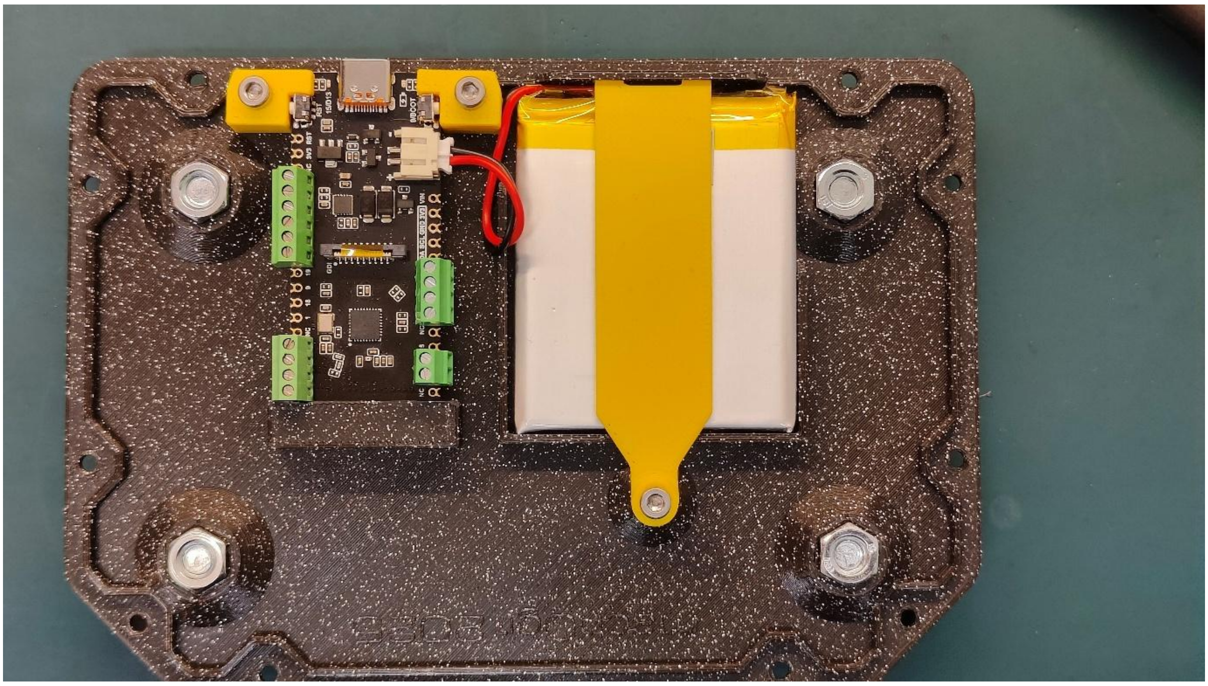
Warning: LiPo batteries can explode or catch fire if mishandled or damaged. They can become unstable and dangerous if punctured or exposed to high temperatures. Handle with care

Place the PCB board into its position by first inserting the back side (antenna) into the slot



Picture 14

Secure the PCB by placing the two little holders as below and secure them by screwing an M3x8mm



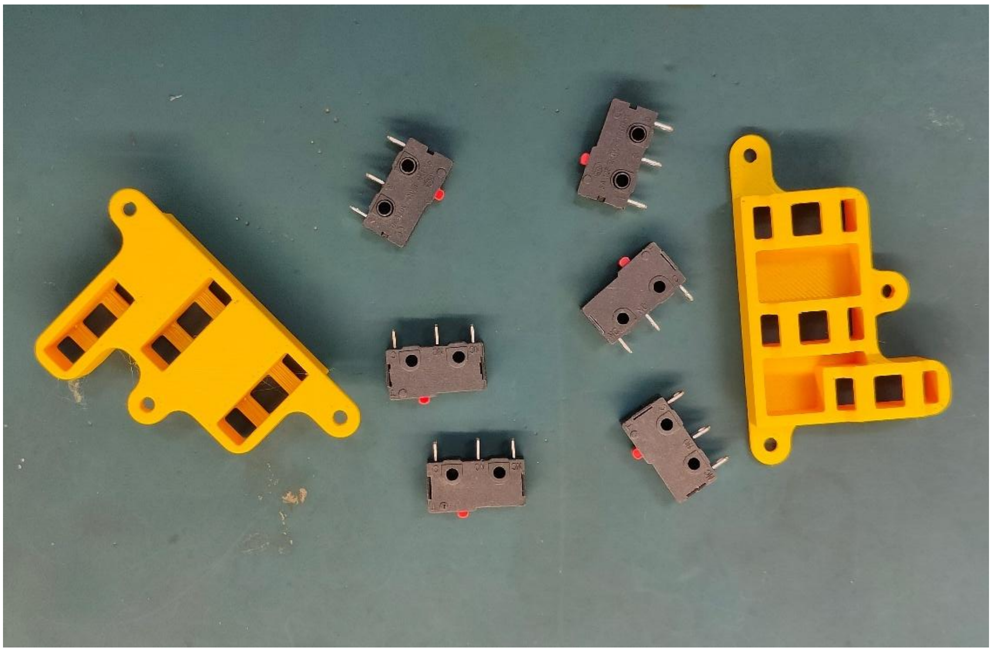
Picture 15

You may connect the battery if you wish.

You can hook it up to with a USB cable on a charger to charge the battery while you complete the rest of the build.

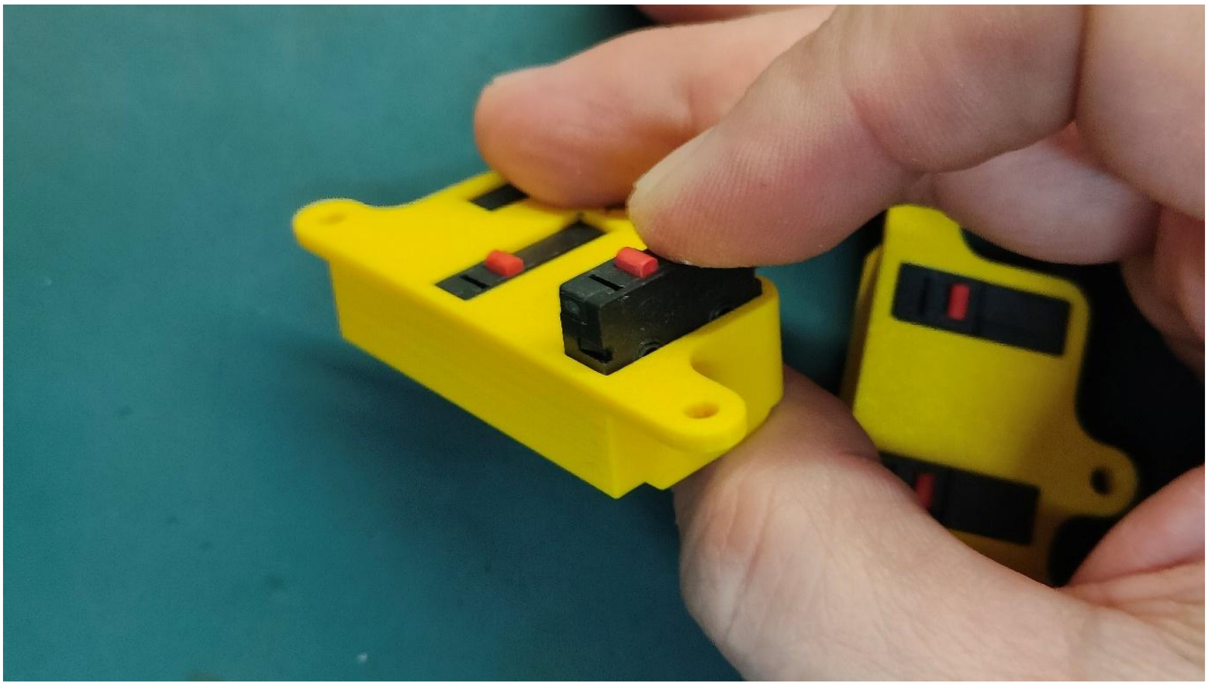
6.3 Assemble top section

To assemble the top section, start with the buttons. Get 6 microswitches and the relevant brackets.



Picture 16

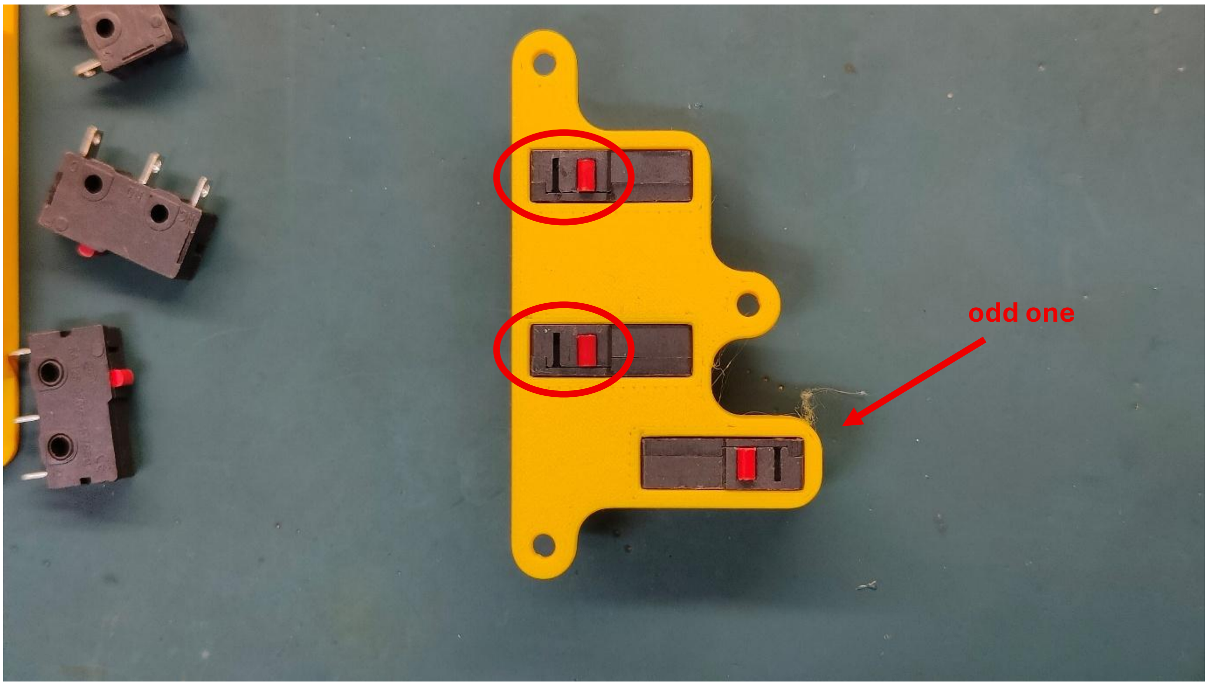
Push in the microswitches in the relevant sockets. Take care to place them in the right way, The clickable side along the long edge. R eference Picture 18



Picture 17

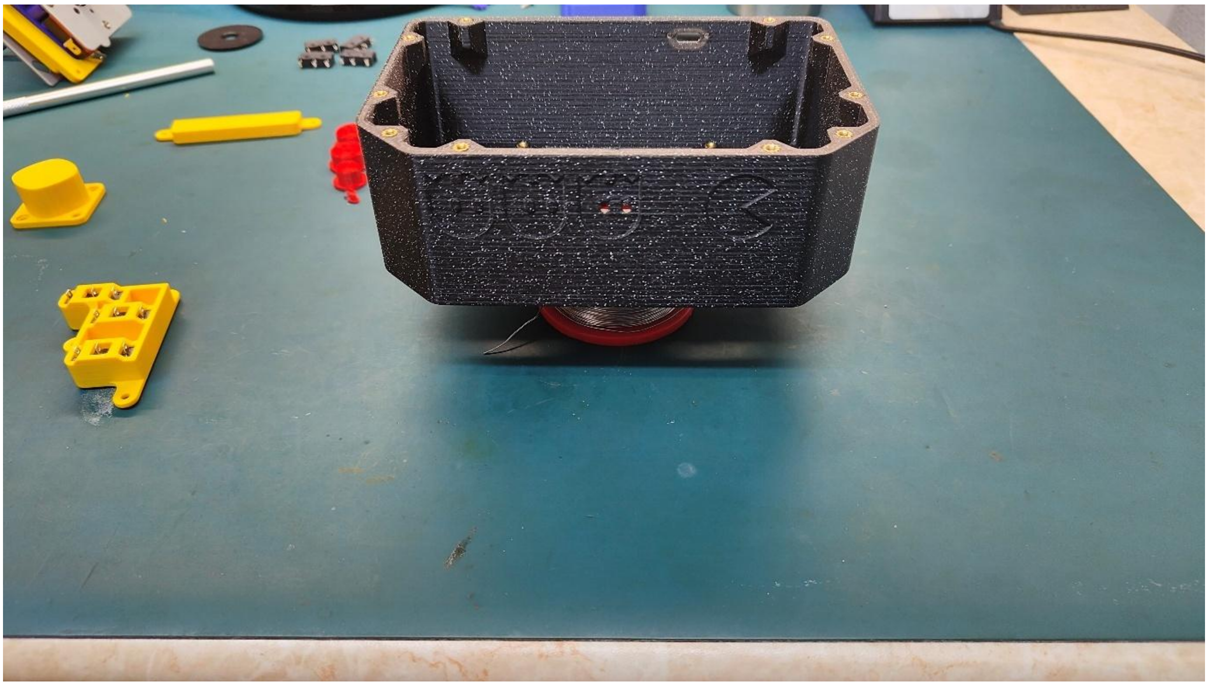
Mind you one of the microswitches, the odd one out, needs to be placed the other way around.

Check Picture 18 below.



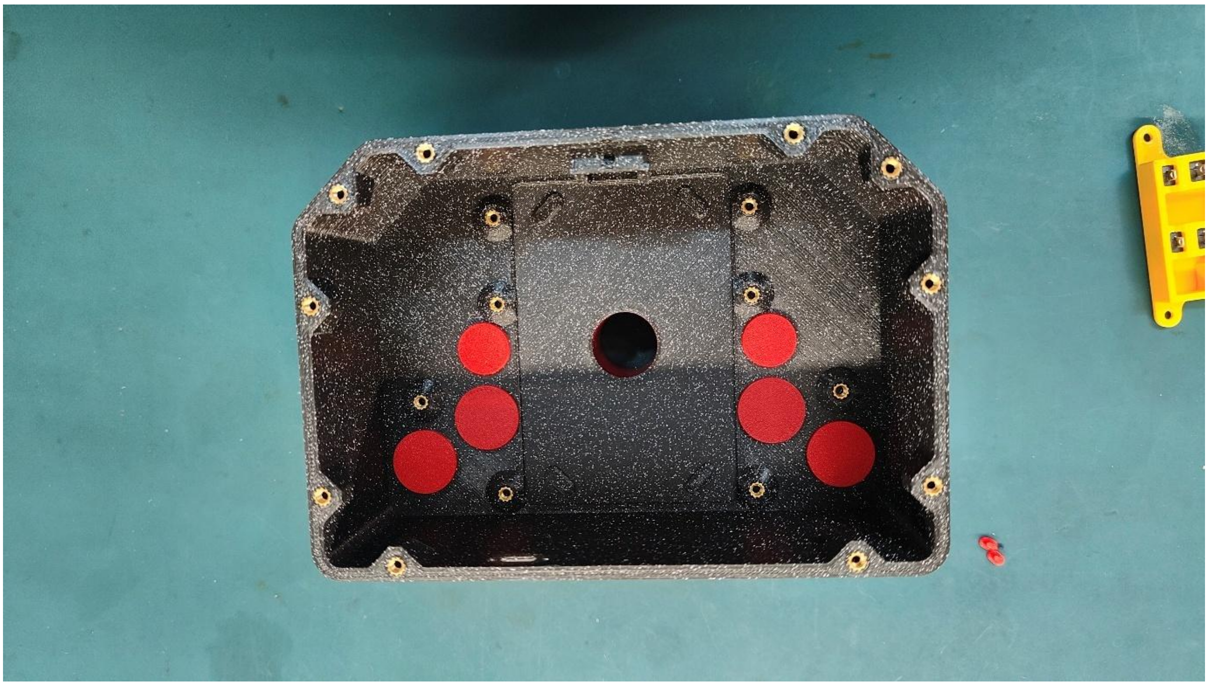
Picture 18

At this point it would be good to find a raiser for your Base box top part. We need somehow to raise the top part a bit to allow us to slide in the buttons. The raiser should not obstruct the button holes. In my case a solder spool placed underneath the top part and at middle did the job just right.



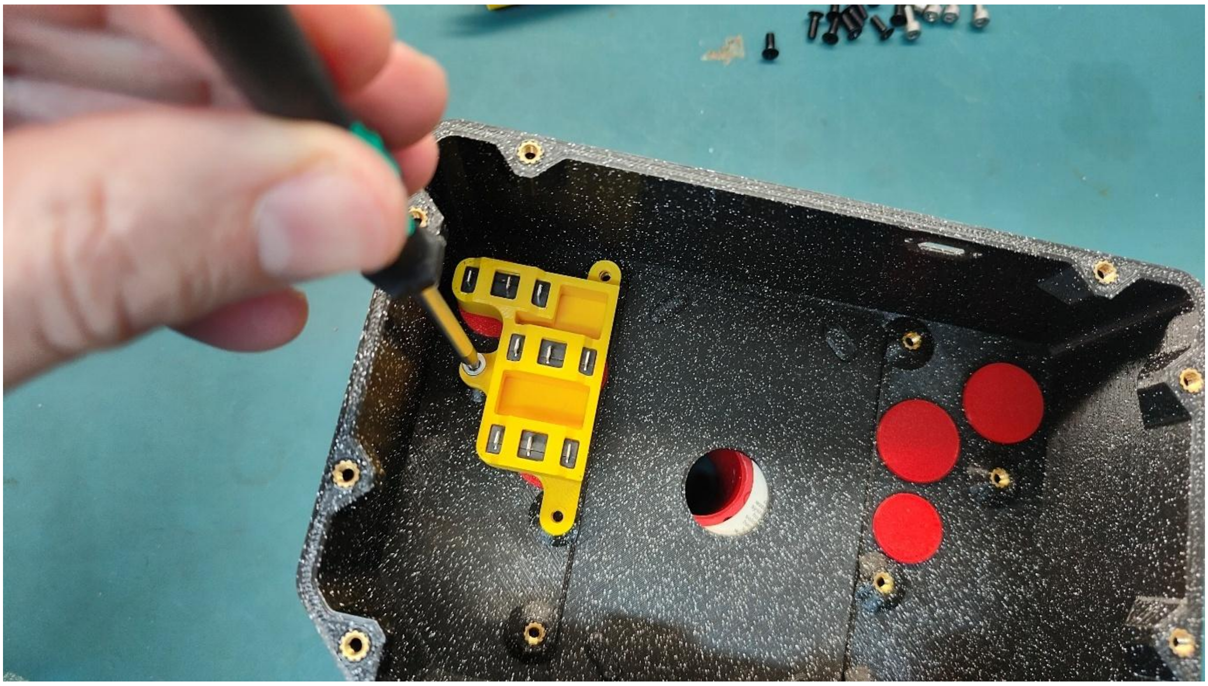
Picture 19

With the box raised, drop all the top 6 buttons in their positions, this is why we need to box raised.



Picture 20

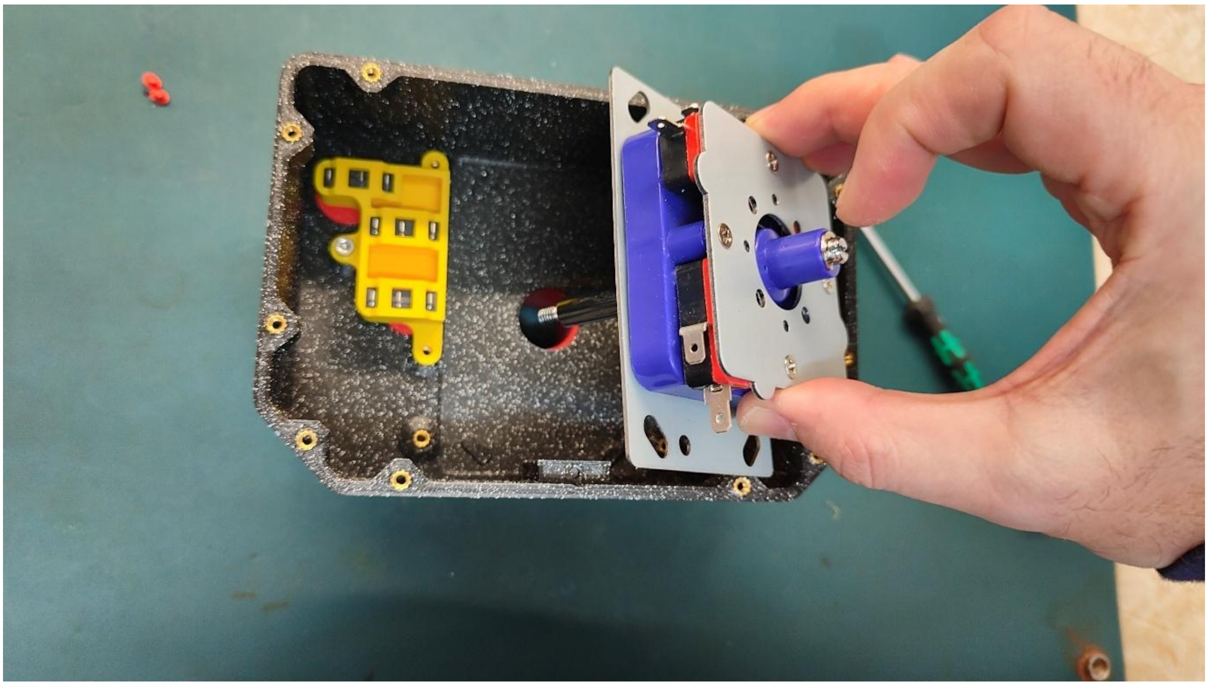
Place the microswitch assembly with the clickable side facing the buttons' bottom side. Align the three screw holes and fix a M3x8mm screw only in the middle hole.



Picture 21

Repeat the same step for the other side. You may want to try and click the nuttons to make sure they are sitting right. They should push in, click and recover nicely.

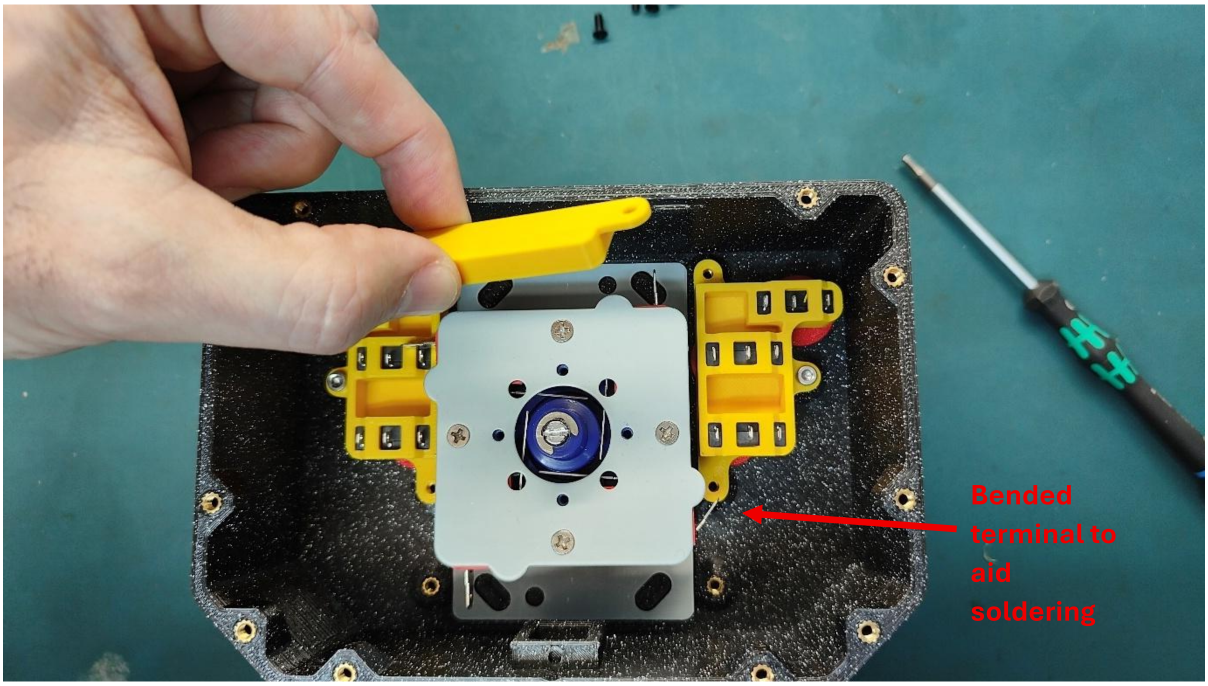
Now glide in the Arcade Joystick Assembly having the ball top removed.



Picture 22

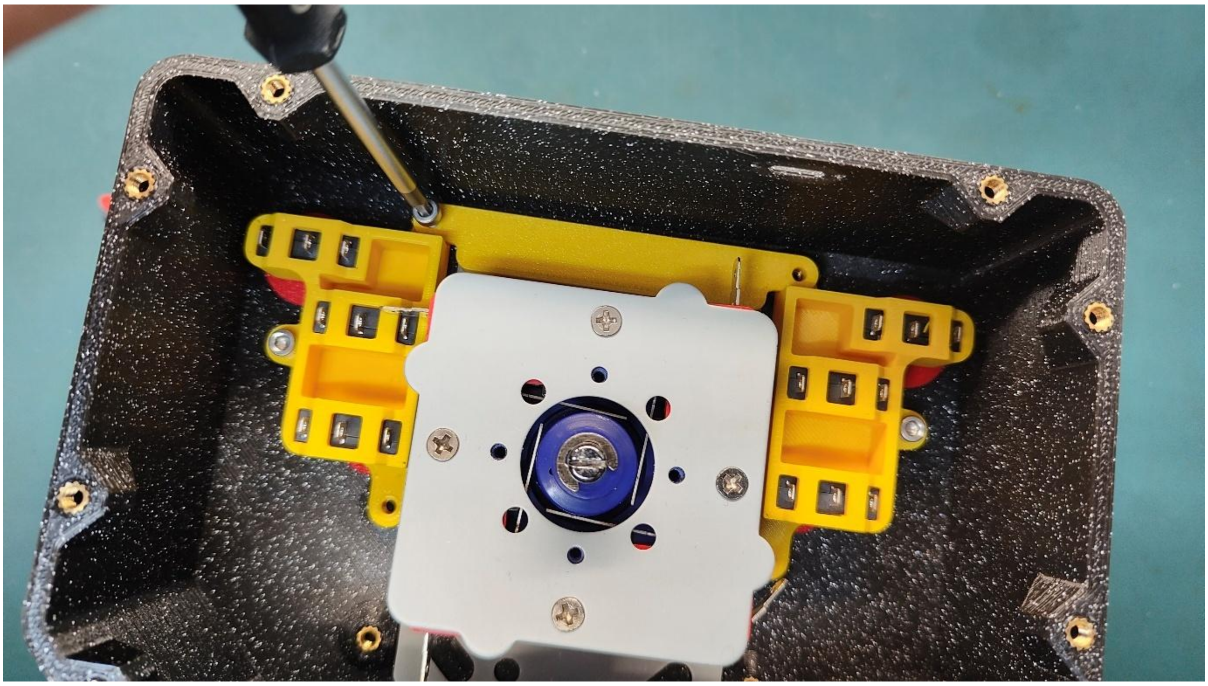
You may want to bend the connection terminals on the Joystick assembly a little bit to the outside which will allow you to solder them easier later on

Now place one of the mounting brackets of the stick as shown below.



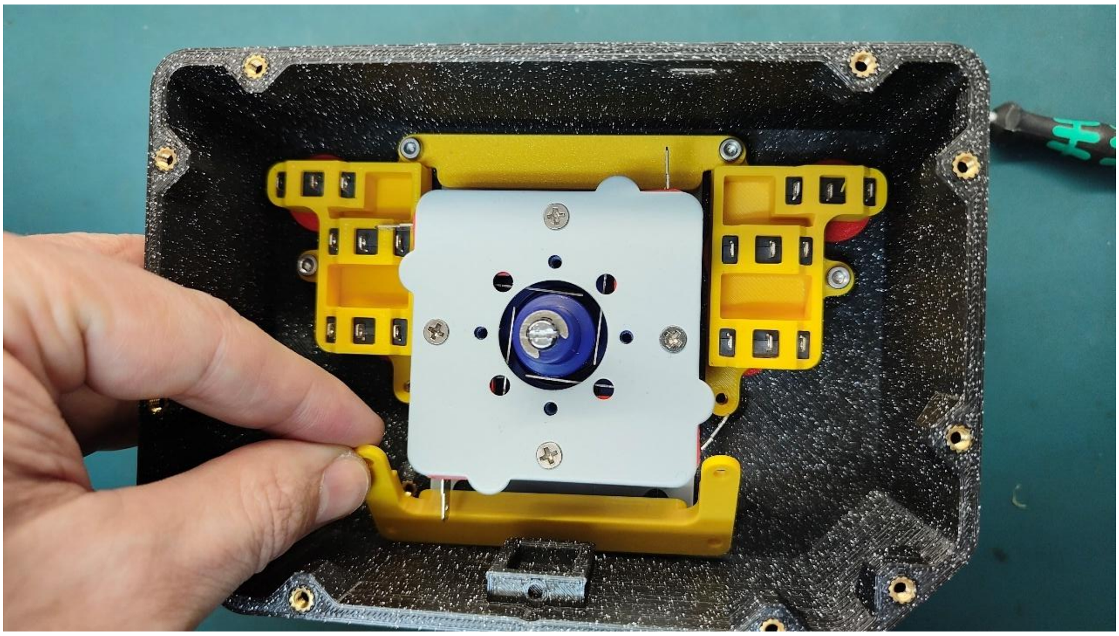
Picture 23

Secure it with a couple of M3x8mm screws.



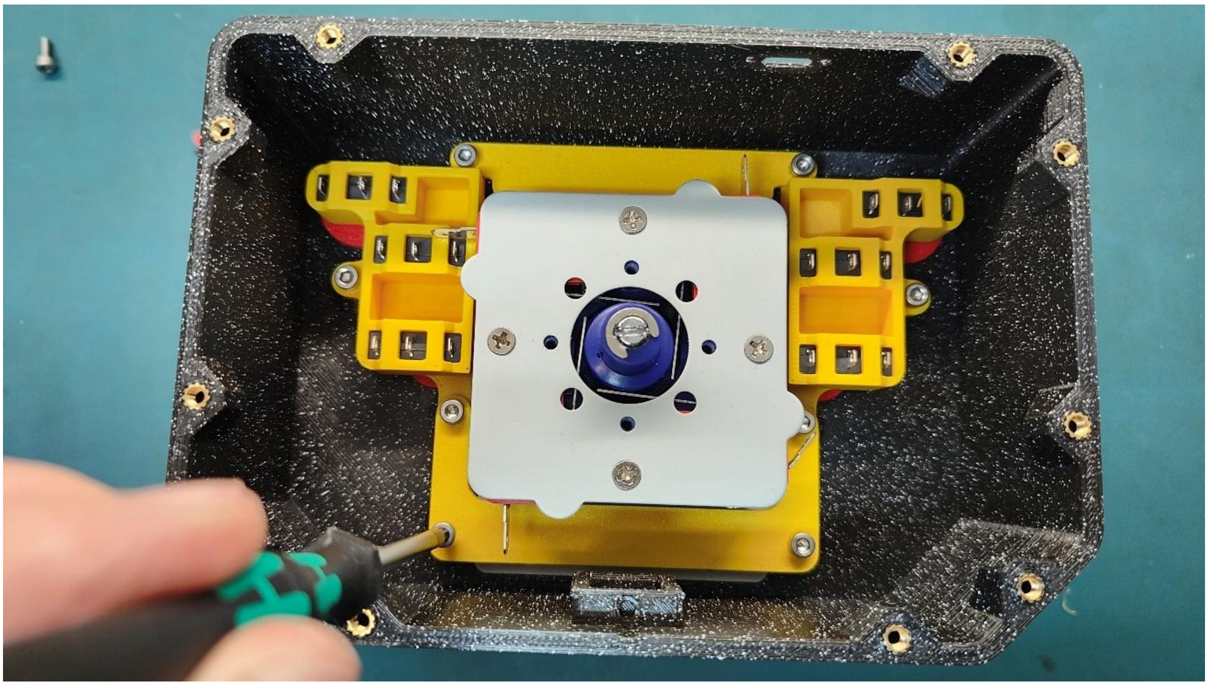
Picture 24

Do the same for the bottom U shaped bracket



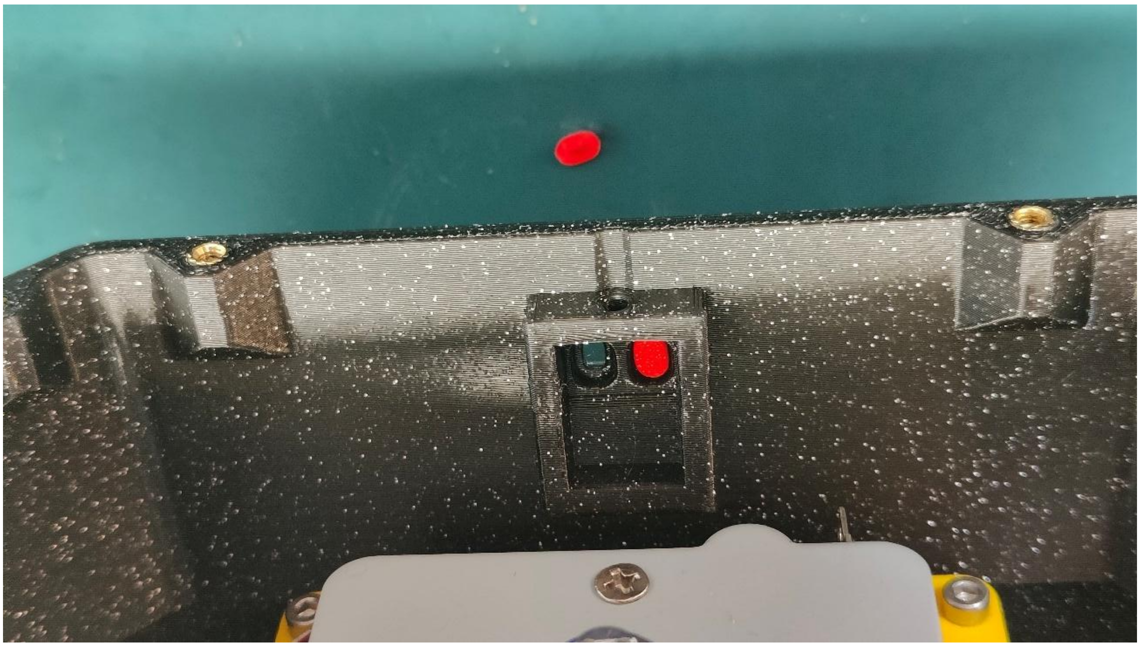
Picture 25

Now secure the bracket with 4x M3x8mm screws



Picture 26

Find the two tiny buttons (the eyes of the pac-man ghost) and slide them in the little holes. You may need to use tweezers to get them to place. You may want to tilt the box so they stay in place for now.

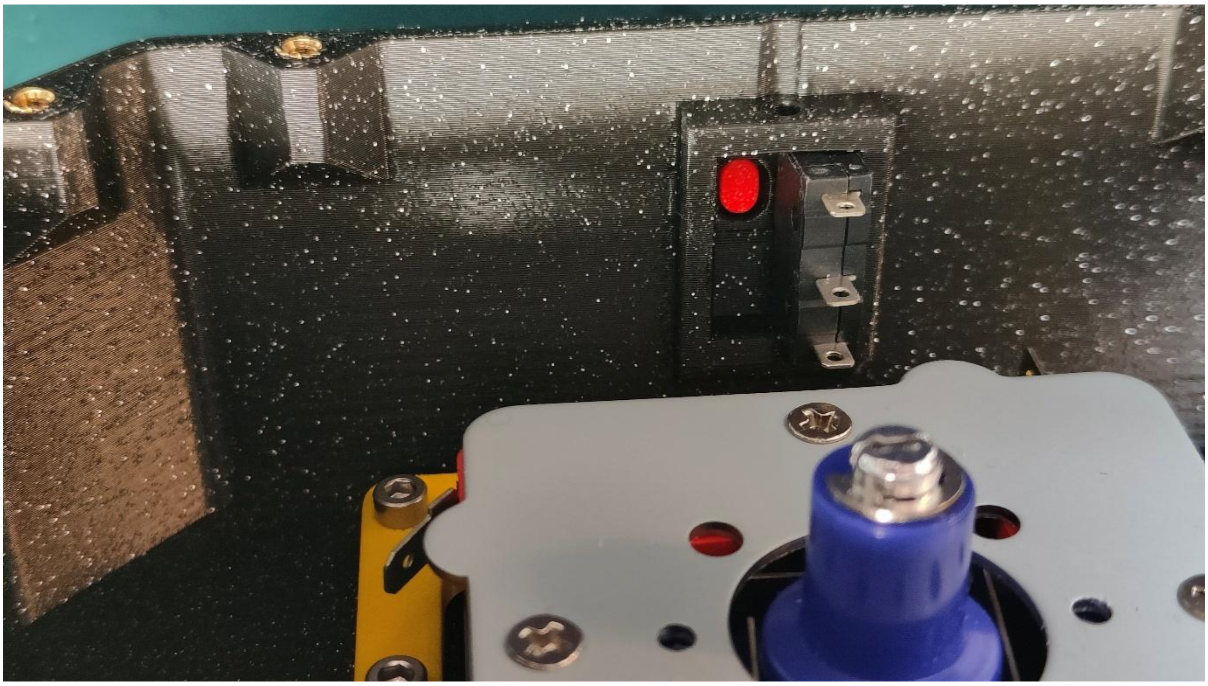


Picture 27

Once you have them in place, pick up the remaining two microswitches and place them in the square section. Make sure the clicky point is going to touch the back side of the tiny button.

Picture 28

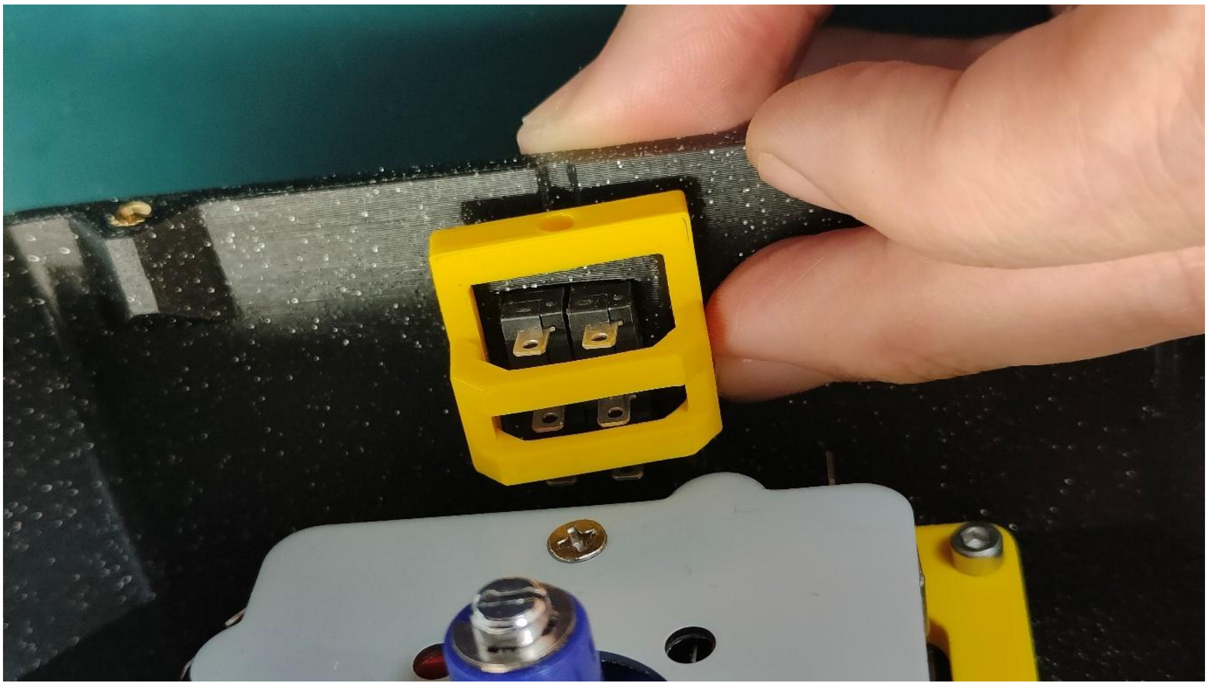
You may need to push it a bit to sit into place, its snug fit by design.



Picture 29

Once in place you can try pushing the button from the outside and make sure the button works correctly. It should click and recover nicely.

Find the microswitch clip and secure both microswitches to place. You will need to place one side first and then push in at the opposite side until you hear the click. Reference the picture 30 below



Picture 30

Both side microswitches should now sit tight and should be clickable from the outside with no issues. The bracket clip should hold them firmly into place. If for any reason they are not, there is an **optional** hole to insert an M3x**6mm** screw. This **optional** screw is **shorter** then the ones of the project and is not in the BOM. You won't need it unless there is slack or microswitches sit loose.

Picture 31 – Optionally securing the bracket clip if loose

Turn the unit upside down and place the stick ring cover that came with the joystick.

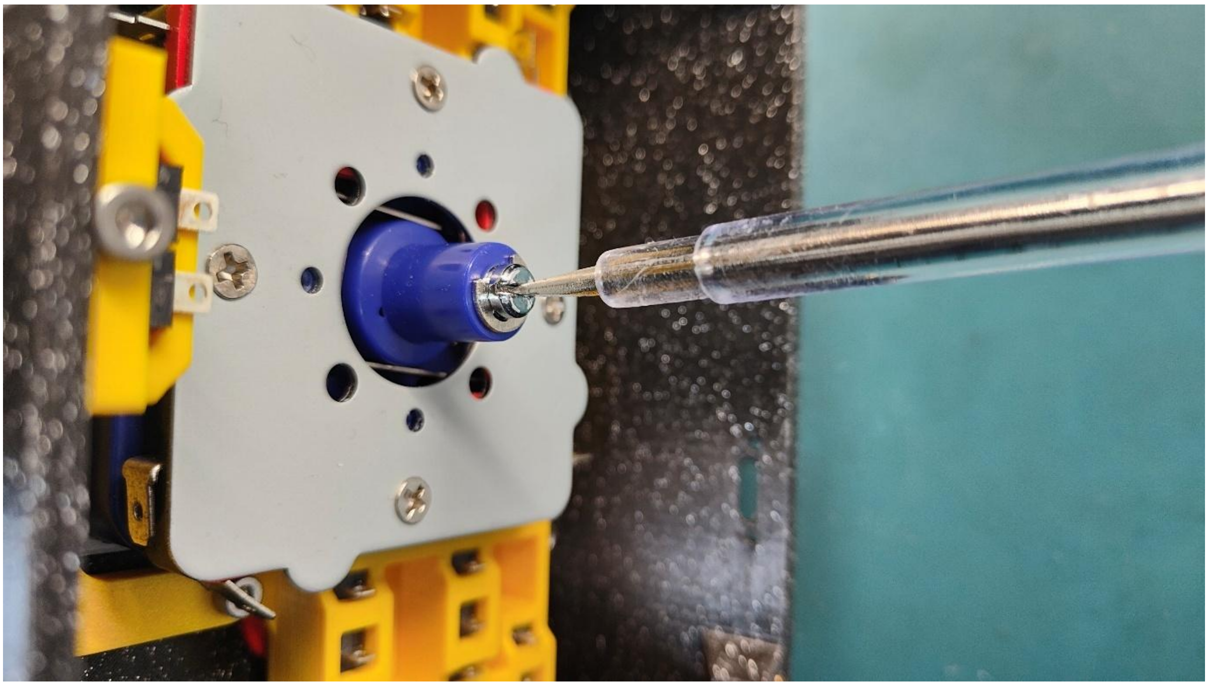
Picture 32

Slide in the stick handle in the top section

Picture 33

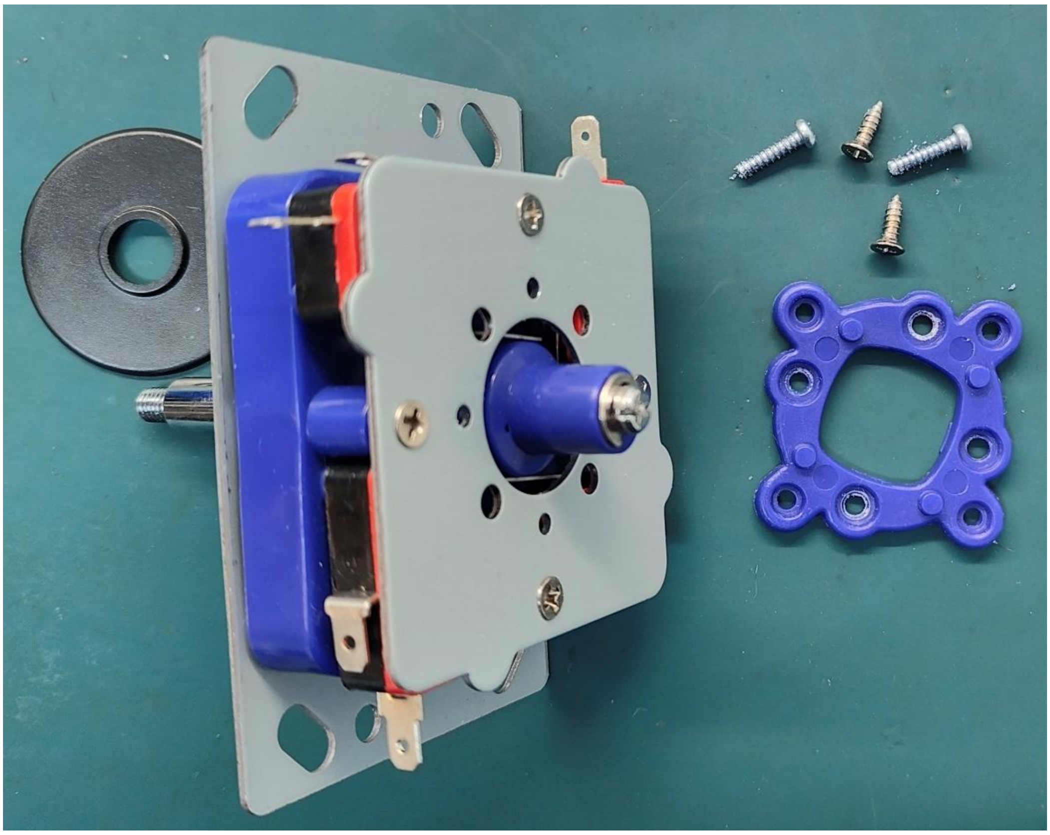
We need to screw in the handle now to the stick. Hold the handle with one hand, use a flat head screwdriver with your other hand and tighten the rod-screw until firmly tight. Reference Pictures 34 and 35 below.

Picture 34

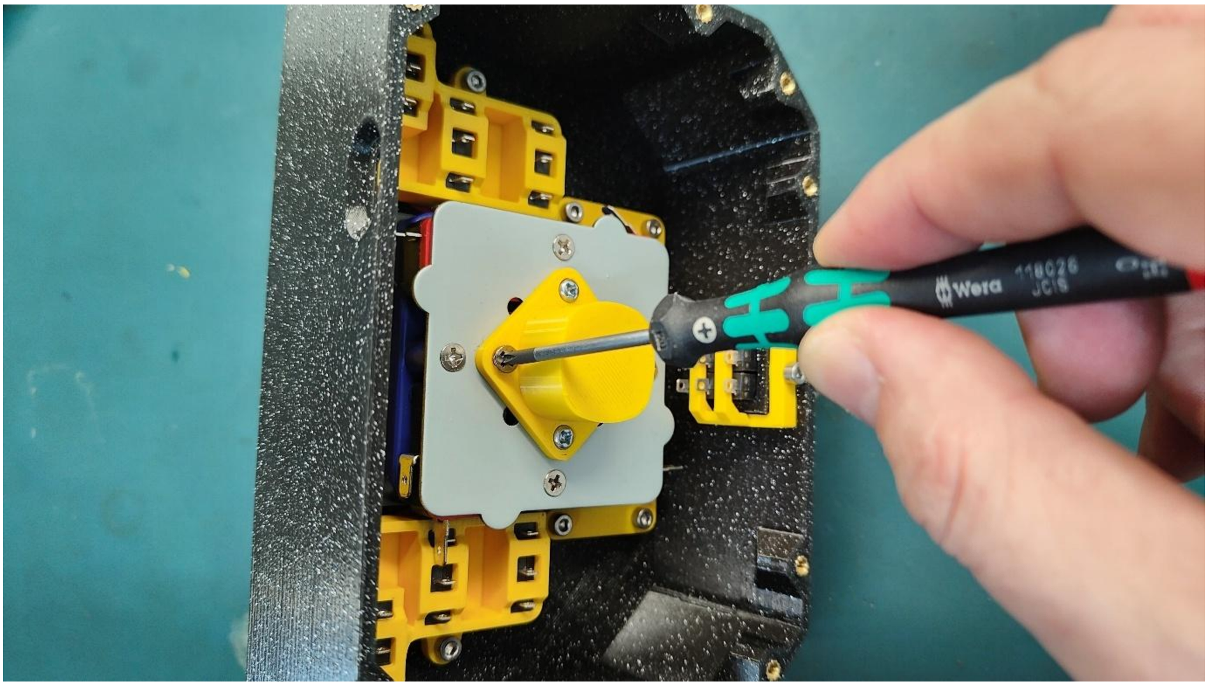


Picture 35

Your stick may have come with a restrictor. You can optionally remove it and place our own restrictor that above all protects the wires underneath.

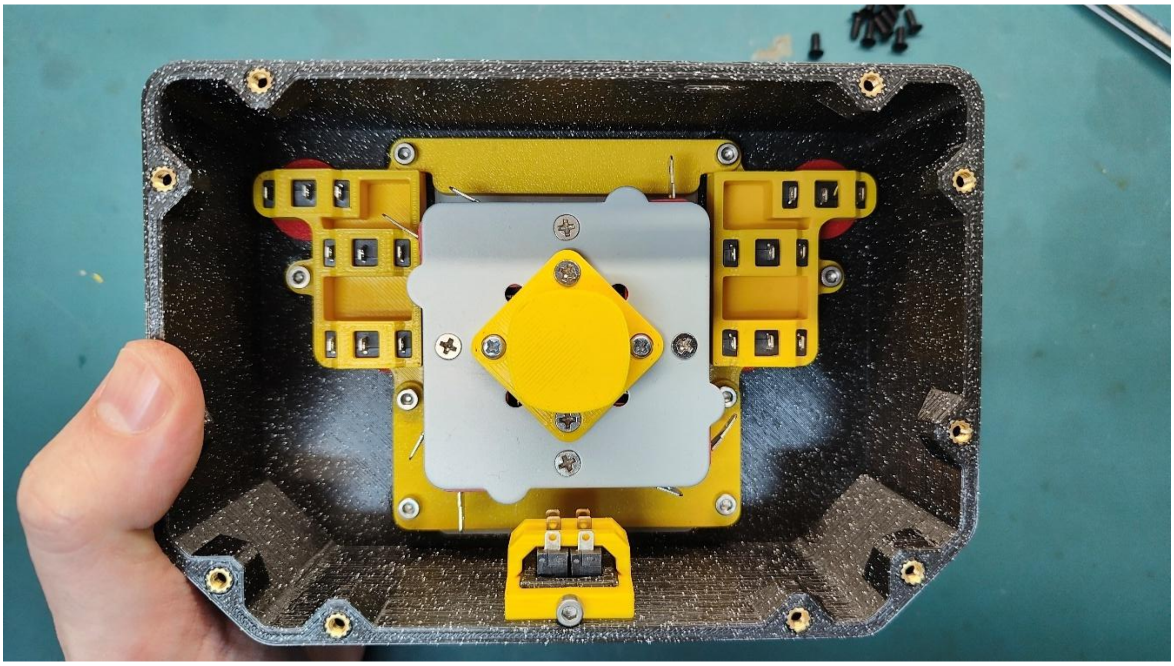


Picture 36



Picture 37 – placing custom restrictor and cable protector

You should now have your top section assembled.



Picture 38

At this stage before proceeding with the cabling, make sure all buttons can get energized (clicked) and recover and that there is no jamming.

6.4 Wiring and soldering

A few words on wiring the Renegade stick.

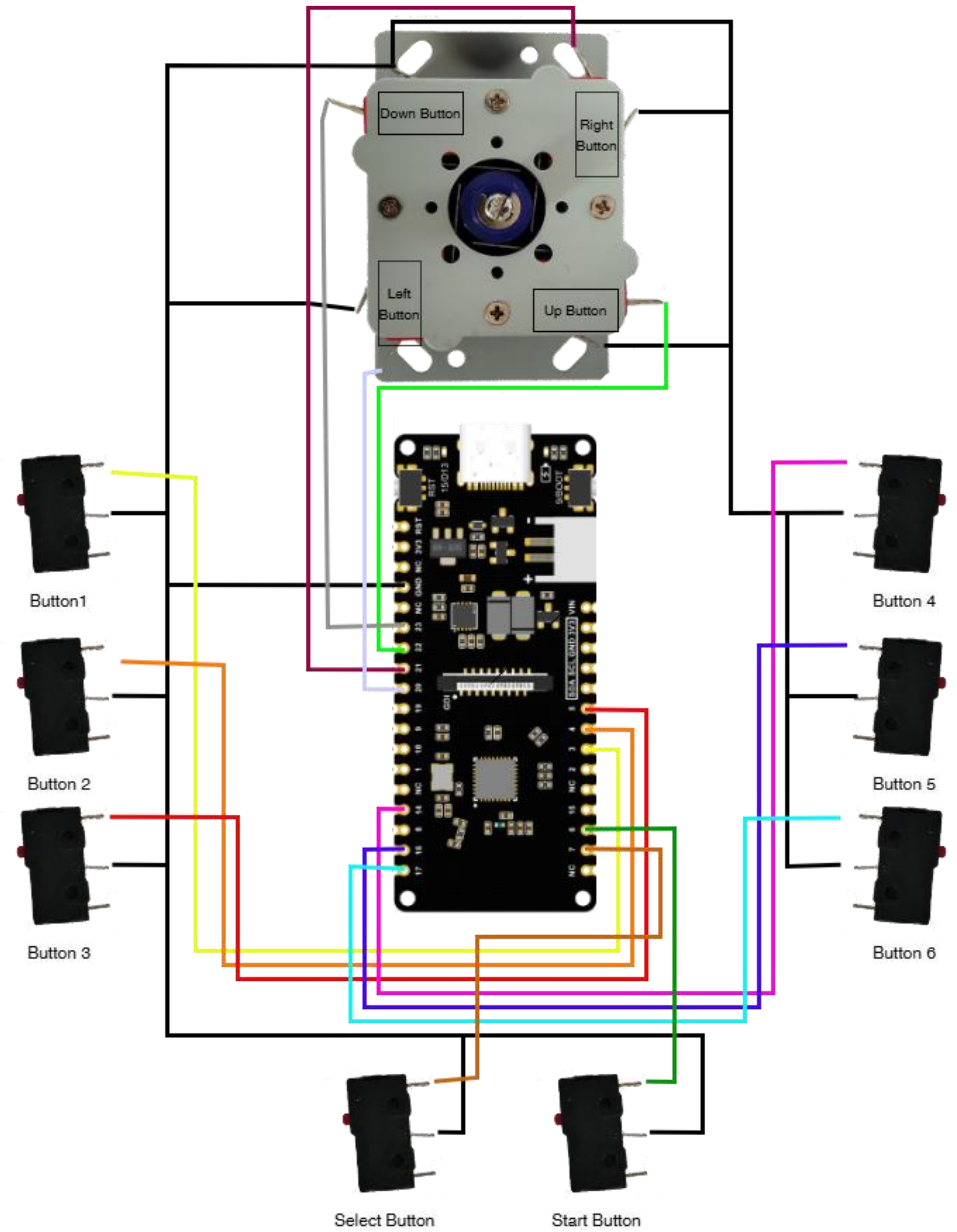
The switches can be wired in many ways. If you are experienced, feel free to skip the guides below and reference the PCB board pinouts directly which you can easily find at the end in the appendix. Below I document the way I wired the joystick which you may find useful.

To start we need to connect one of the terminals of each microswitch (joystick assembly included) to ground (GND). The other terminal is our signal, which needs to get connected to one of the pins on the PCB board.

If your Joystick assembly came with "two terminal" type of switches, that's great, one of them (either) needs to connect to ground. If your assembly came with "three terminals", you need to find which one is the "common". Use a multimeter in continuity mode and find the pair of terminals that beep your multimeter when the button is pressed.

Regarding all the other microswitches you should typically find them in "three terminal" type. For these, the "common" terminal is usually the one closest to the click button while the "normally open (NO) terminal usually positioned in the middle. We need to use the "common" and the "normally open" terminals. Please try to test the microswitches you souced with a multimeter and identify the terminals you need to use. Again, it should be a pair that "close circuit" when the button is pressed. Study the images below and you should identify the terminals I'm using, they should be the same for your case too.

Next you will find an example connection diagram to an ESP32-C DFR1075 board. If you sourced a different board reference the matrix displayed earlier in section 6.1 or check the last pages of the document in appendix where there are all board connection diagrams for easy reference.



Picture 39 - connection diagram to an ESP32-C DFR1075

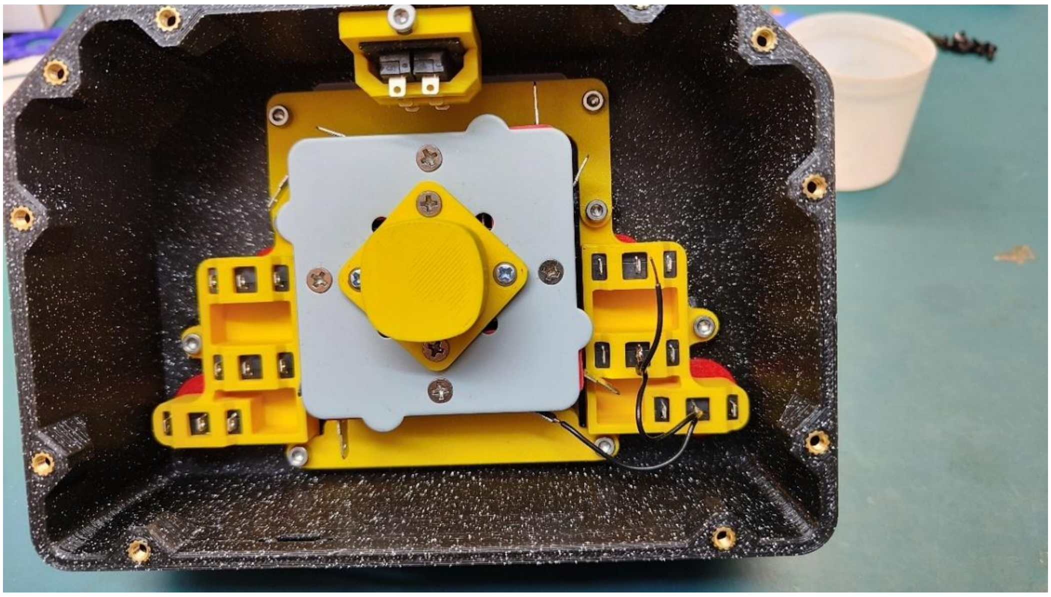
Depending on how you have your items laid down on your table, make sure you're looking the joystick assembly the correct way and not 180° upside down. Mind you however that pushing the level stick up for example activates the bottom microswitch and similarly this logic applies to all other directions

Now, there are **two ways** you can wire this project up. **Soldering** the wires directly on microswitch terminals or using **“Quick Disconnect Female” connectors**. Each has their pros and cons, I tried both ways, both take time but I think the easiest way and more robust way for the average user would be to solder.
When soldering, watch out for the hot part of the iron—not just the tip! It’s easy to bump into things in tight spaces. You don’t want to accidentally touch and melt your casing or even a cable sleeve.

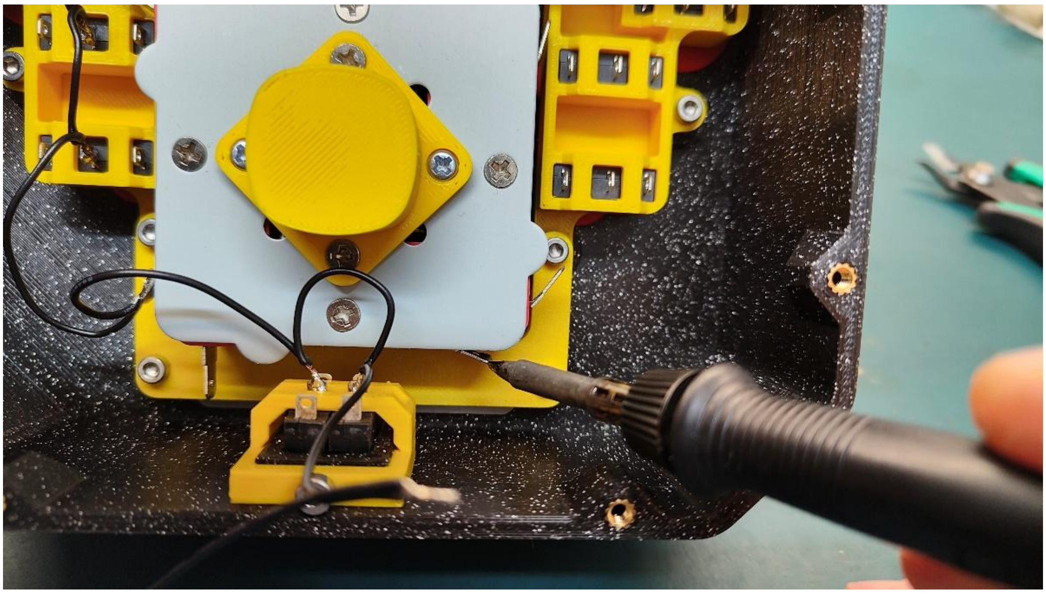
Please watch out how you move your hot iron inside the confined space of the box.

Below is my take on how I wired the whole thing.

An easy way is to pick the middle terminal from all the microswitches and daisy chain them all to a common wire, the ground(GND).



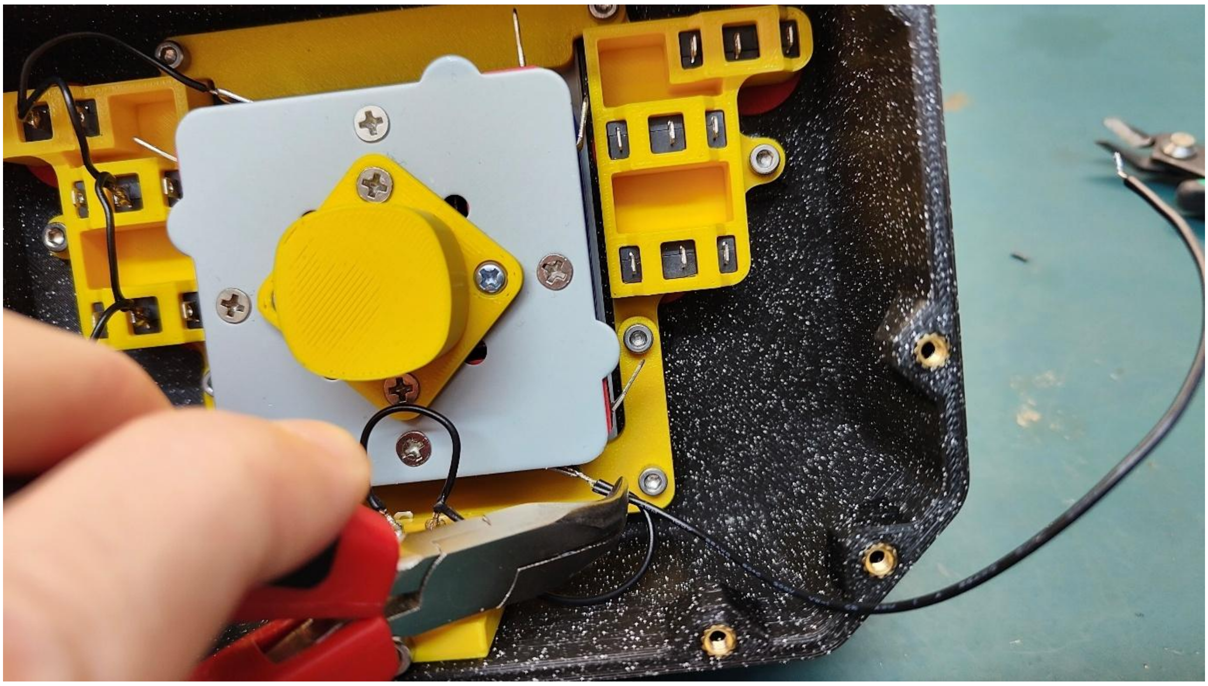
Picture 40 – daisy chaining the GND



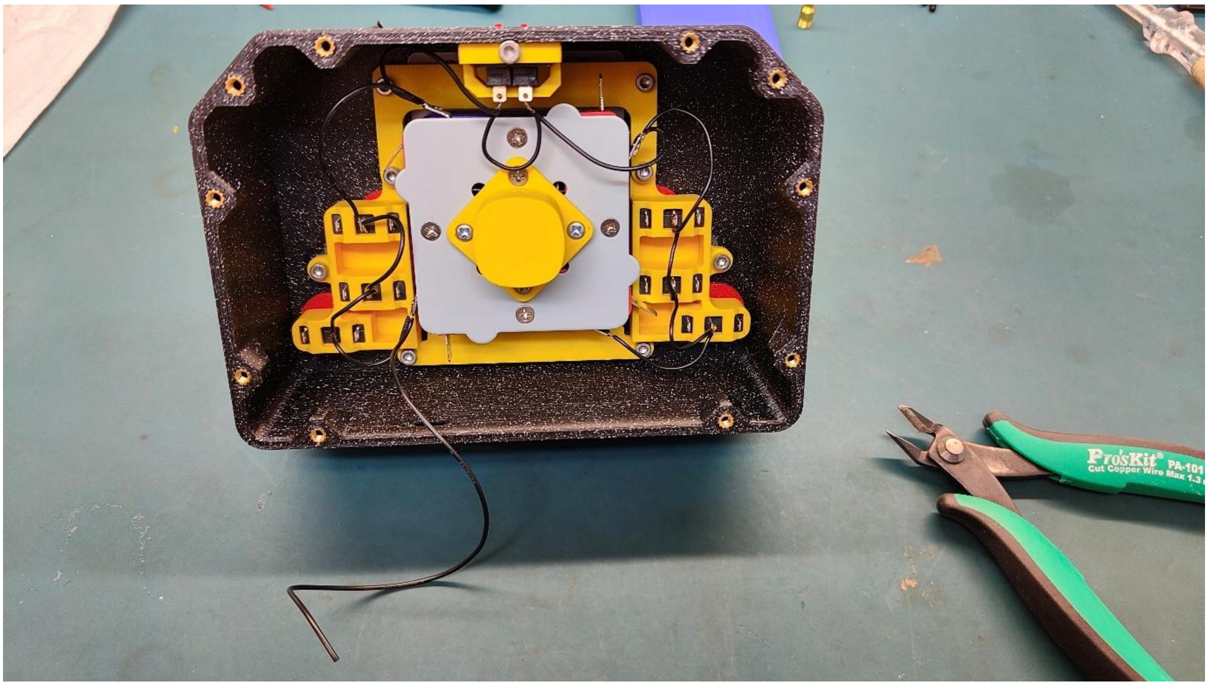
Picture 41 – more daisy chains of GND

You may want to melt some solder on the terminal first. Solder the wire strands end tip and lastly bring the two parts together and touch them with the hot iron tip.
In some cases a tool can help keep your wires while you reach down to confined spaces.

Picture 42



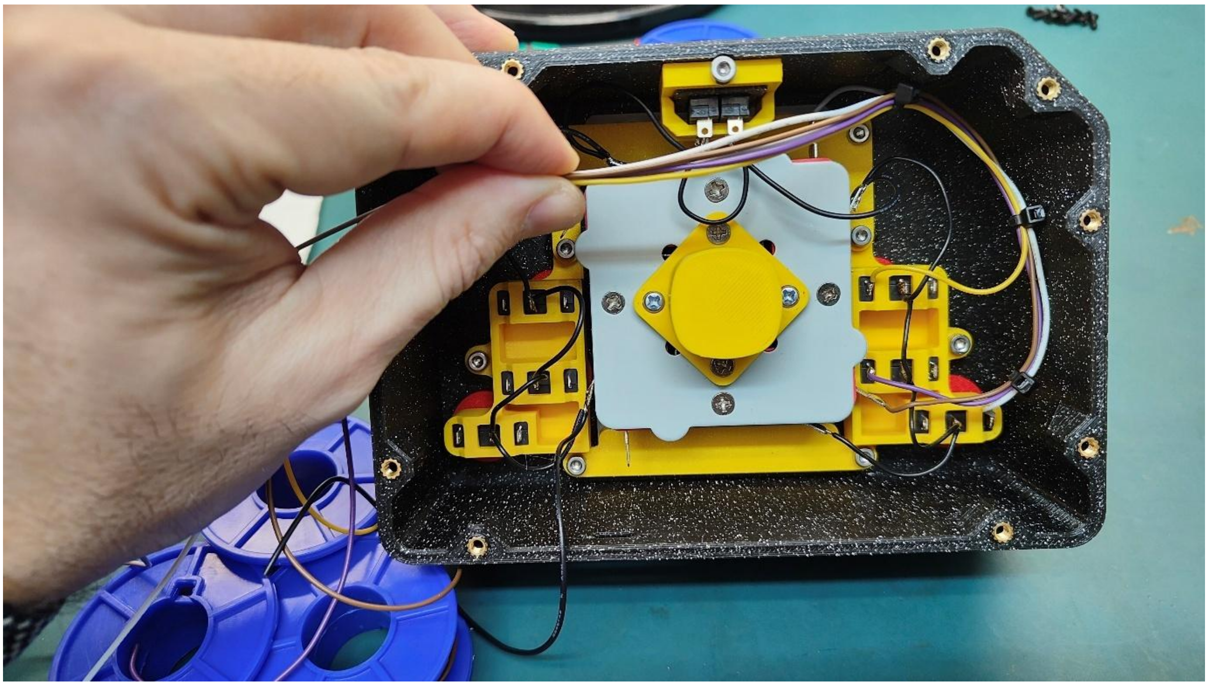
Picture 43



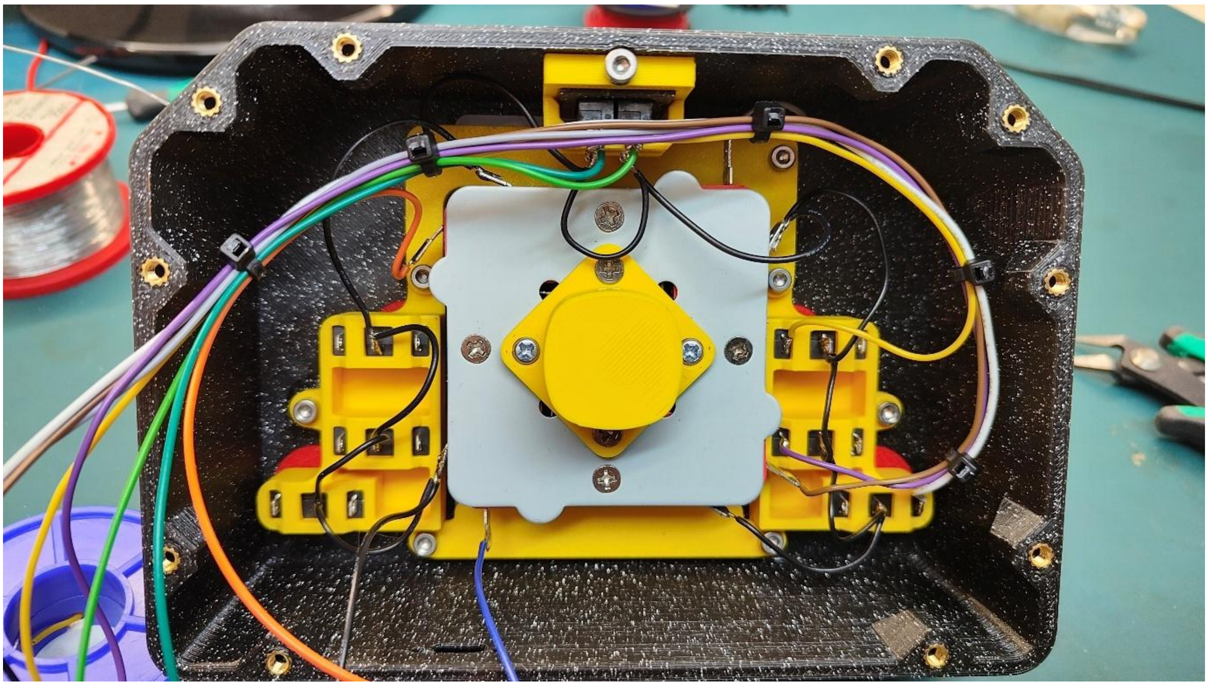
Picture 44 – GND wiring complete

Completing the GND chain, move to the other terminal of the switches. I did start the wiring from bottom right section as seen below and moved in a circular way using cable ties to keep everything clean.

If you want to precut your cables, a length of 350mm will do just fine even for the longer one.

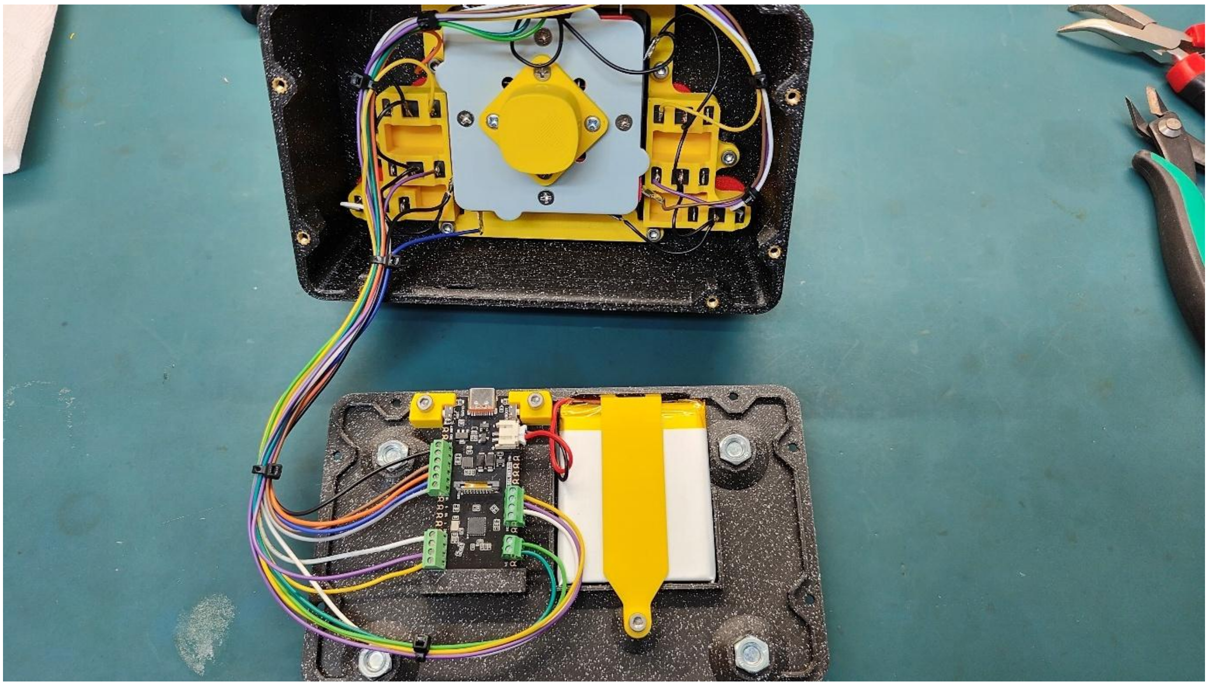


Picture 45



Picture 46

Working your way around all the microswitches with patience you'll end up with something like below.



Picture 47

If you have already programmed the board then you can go ahead and close the box using the countersunk screws. If not, you have to *flash* the board (visit gitgub page) now as you may need access to the PCB board side buttons for programming.

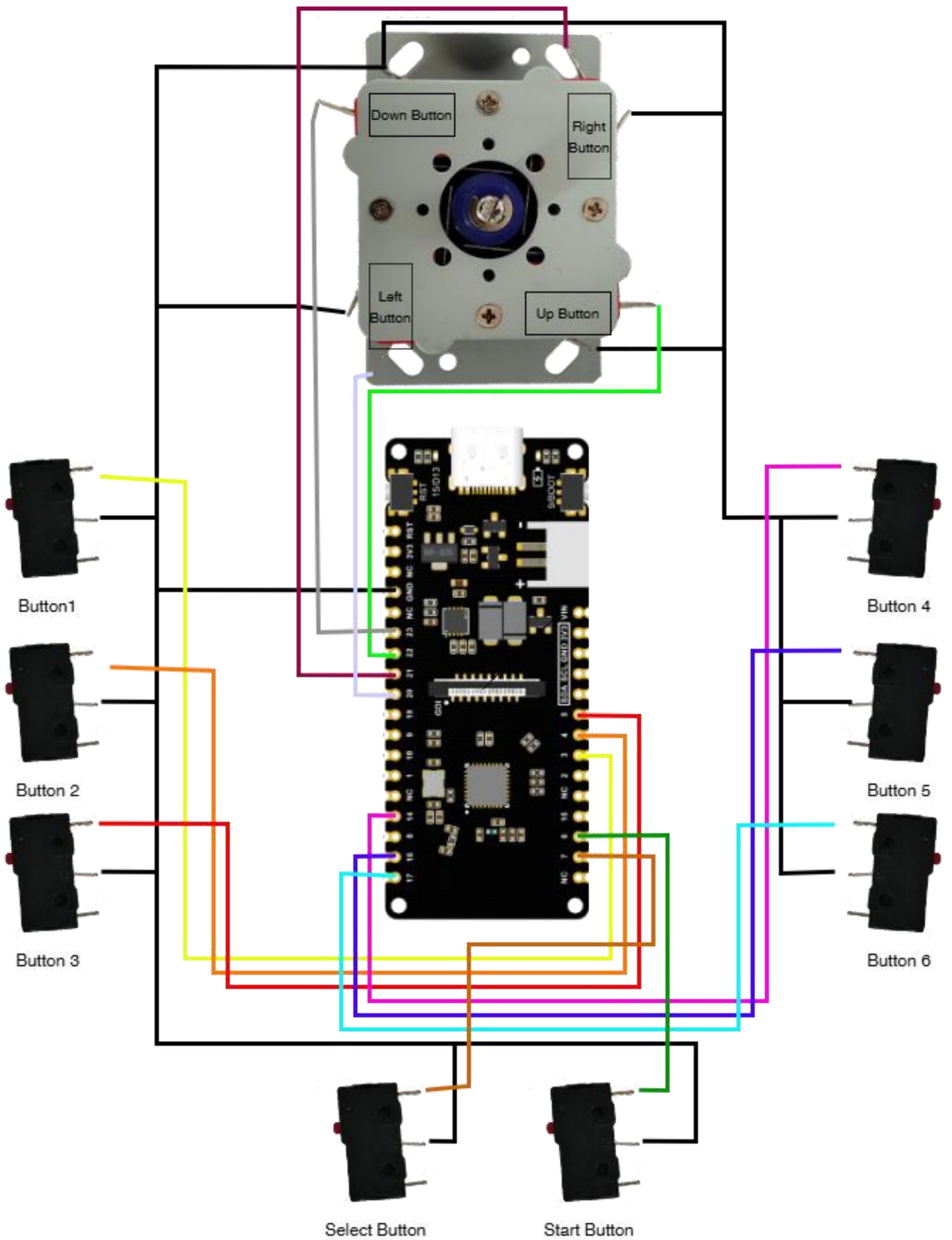
Picture 48

Awesome job! You've built your Renegade Stick. Now head over to the GitHub page to flash the board and start playing!

7. Flashing, powering up and pairing with a host

This document serves as an assembly guide so to give life to the Renegade Stick you'll have to head over to the Renegade Stick on Gitgub page.
You'll find all required information there for flashing, powering up, pairing with a host, troubleshooting and more.

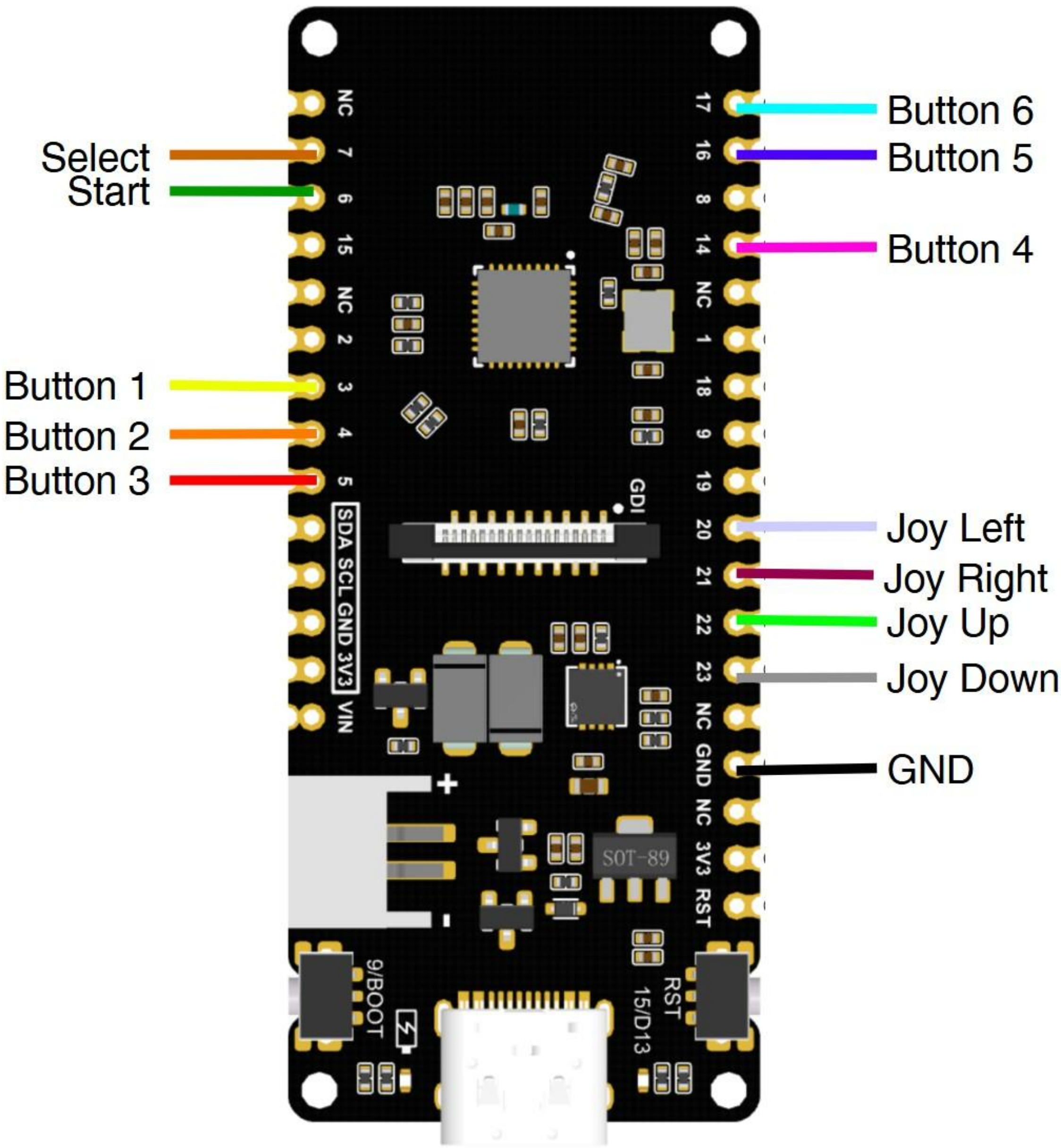
8. Appendix: Quick reference figures, drawings and diagrams



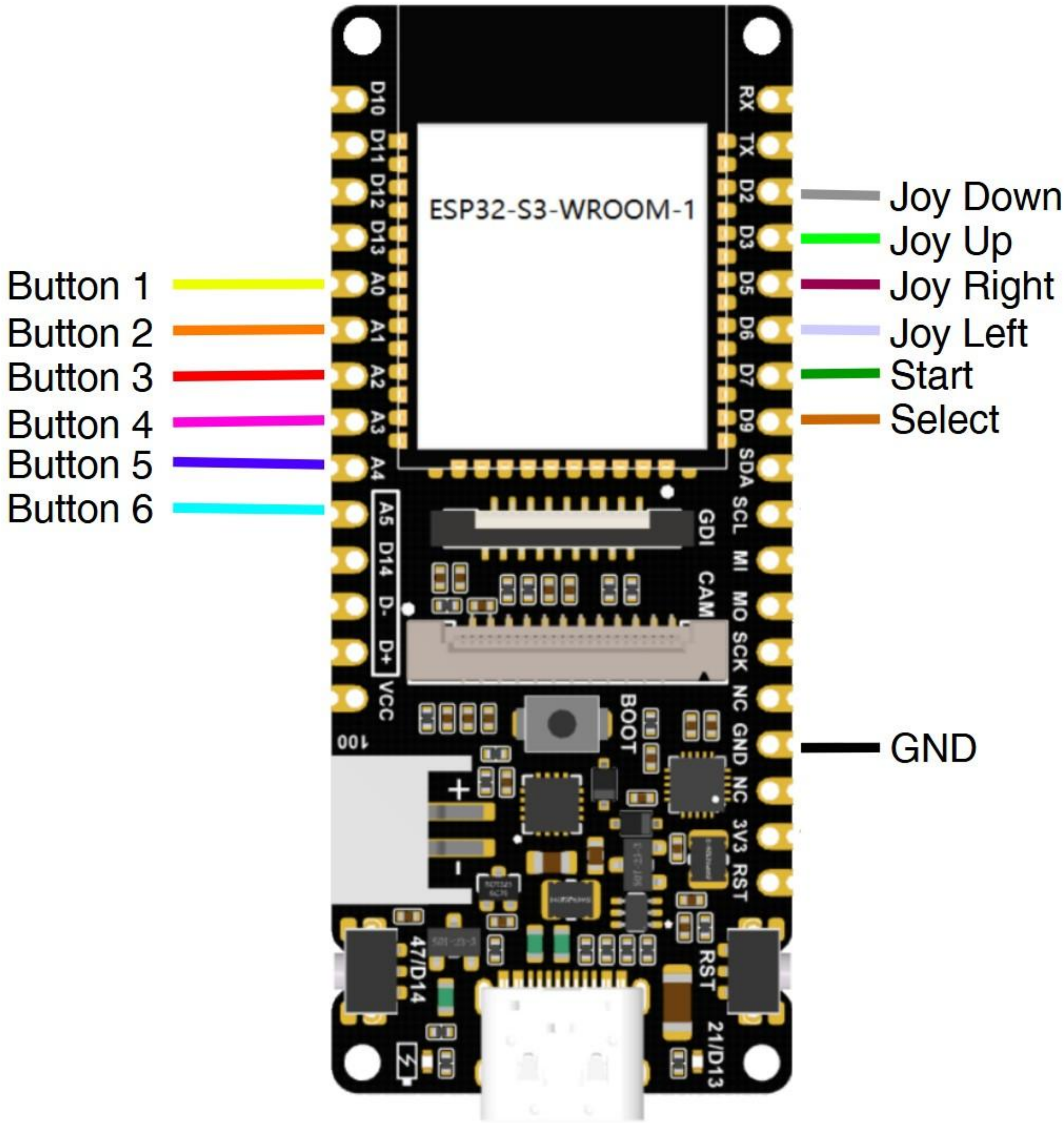
Picture 49 - Wiring Diagram for the ESP32-C6

	ESP32-C6 DFR1075	ESP32-S3 DFR1145	ESP32-E(N16R2) DFR1139
BUTTON_1	GPIO_NUM_3 (A2)	GPIO_NUM_4 (A0)	GPIO_NUM_22 (SCL)
BUTTON_2	GPIO_NUM_4 (A3)	GPIO_NUM_5 (A1)	GPIO_NUM_21 (SDA)
BUTTON_3	GPIO_NUM_5 (A4)	GPIO_NUM_6 (A2)	GPIO_NUM_15 (A4)
BUTTON_4	GPIO_NUM_14 (D3)	GPIO_NUM_8 (A3)	GPIO_NUM_12 (D13)
BUTTON_5	GPIO_NUM_16 (TX)	GPIO_NUM_10 (A4)	GPIO_NUM_4 (D12)
BUTTON_6	GPIO_NUM_17 (RX)	GPIO_NUM_11 (A5)	GPIO_NUM_17 (D10)
START_BUTTON	GPIO_NUM_6 (D12)	GPIO_NUM_9 (D7)	GPIO_NUM_13 (D7)
SELECT_BUTTON	GPIO_NUM_7 (D11)	GPIO_NUM_0 (D9)	GPIO_NUM_14 (D6)
DPAD_L	GPIO_NUM_20 (SCL)	GPIO_NUM_18 (D6)	GPIO_NUM_0 (D5)
DPAD_R	GPIO_NUM_21 (MI)	GPIO_NUM_7 (D5)	GPIO_NUM_26 (D3)
DPAD_UP	GPIO_NUM_22 (MO)	GPIO_NUM_38 (D3)	GPIO_NUM_25 (D2)
DPAD_DOWN	GPIO_NUM_23 (SCK)	GPIO_NUM_3 (D2)	GPIO_NUM_1 (TX)

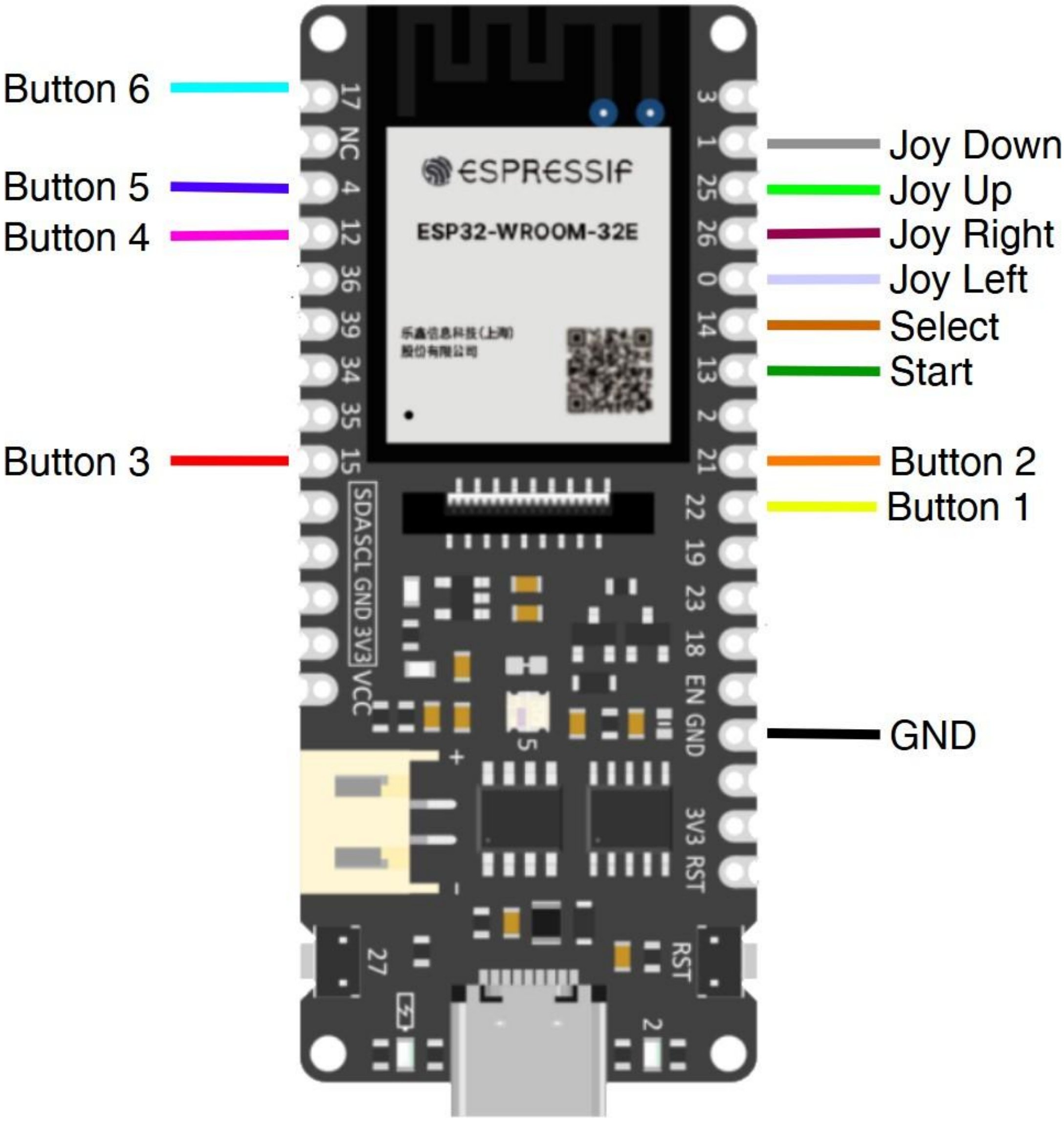
PCB Board proposed connection pinouts



Picture 50 - ESP32-C6 DFR1075 proposed connection diagram



Picture 51 - ESP32-S3 DFR1145 proposed connection diagram



Picture 52 - ESP32-E (N16R2) DFR1139 proposed connection diagram